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Optimising Weak Reference Processing in the JVM Z Garbage Collector

Reference Pipeline Optimisations and an Exploratory

Weak-Field Mechanism for the JVM



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Abstract

Reference processing can be a critical component of Java garbage collection latency, particularly in workloads with large numbers of weakly reachable objects. This thesis investigates how the Z Garbage Collector (ZGC) can reduce overhead when processing `WeakReference` objects that are not associated with a `ReferenceQueue`. Three orthogonal implementation strategies are developed in an OpenJDK research fork: a queue-less fast path that skips enqueue handling, a cache-friendlier dynamic-array representation for discovered references, and an optimised clear path that removes unnecessary per-reference work.

To evaluate these changes, a controlled benchmarking framework was constructed with repeated runs, CPU isolation, memory monitoring, and GC-log analysis. The evaluation compares all combinations of the three optimisation strategies against a baseline configuration using both single-object stress and multi-object phased-release workloads.

In addition to `WeakReference`-based optimisations, the thesis introduces an exploratory weak-field mechanism via a Java field annotation and corresponding VM support. This allows a representation-level comparison between object-based weak references and field-based weak semantics under the same experimental setup.

To be continued...

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1. Introduction

Java is one of the most widely used programming languages in enterprise servers [1], where applications routinely manage heaps ranging from tens to hundreds of gigabytes. In such environments, the performance of the Java Virtual Machine’s (JVM) garbage collector (GC) directly affects application latency and throughput. The Z Garbage Collector (ZGC) addresses this challenge by performing nearly all garbage collection work concurrently with the application, thereby achieving millisecond-scale pause times regardless of heap size [2].

However, one aspect of ZGC that has received comparatively little optimisation attention is the processing of *weak references*. In Java, the class `WeakReference` provides weak semantics, allowing applications to retain references to objects without preventing those objects from being collected [3, 4]. When the garbage collector determines that an object is only reachable through a `WeakReference`, i.e. no longer strongly reachable, it clears the reference and, if the reference is associated with a `ReferenceQueue`, enqueues it for application-level processing [5, 6]. This enqueueing step serves as a garbage collector callback, notifying the application that the `WeakReference` has been cleared. Weak semantics, both with and without such callbacks, are used in a range of programming patterns and data structures, for example the Java library `WeakHashMap` [7].

The current ZGC reference-processing pipeline treats all weak references uniformly: discovered references are linked into an intrusive linked list, processed, and then handed to a Java-level daemon thread (the `ReferenceHandler`) for enqueueing [8, 5]. This design imposes disproportionate overhead on weak references that are not registered with a `ReferenceQueue`. Such references are used solely for weak semantics and therefore do not require the callback mechanism provided by the queue. Nevertheless, the `WeakReference` object still needs to go through the entire processing pipeline, even though the application has no need for enqueueing.

This inefficiency has previously been identified in the OpenJDK bug tracker [9], which notes the unnecessary traversal of the pending list for references without a `ReferenceQueue`. However, as of April 2026, no implementation addressing this issue has been integrated into the JDK. Building on that observation, this thesis investigates whether targeted modifications to ZGC’s reference-processing pipeline can reduce the overhead of reference processing. It explores three orthogonal optimisation approaches for `WeakReference`-based processing: skipping the enqueue path for

references without queues, replacing the intrusive linked list with a dynamic array, and optimising the codepath of clearing references. In addition, it examines a weak-field mechanism as an alternative implementation of weak semantics that avoids `WeakReference` objects entirely.

1.1 Motivation

Weak references, unlike regular references, require explicit processing by the JVM garbage collector [8, 10]. Each discovered weak reference must be individually evaluated, its referent field cleared if the referent is unreachable, and the reference potentially enqueued. This per-reference processing cost scales linearly with the number of weak references, making it a bottleneck in applications that allocate a lot of them.

The problem is amplified by a mismatch between the API design and its implementation. Java’s `WeakReference` API provides an optional `ReferenceQueue` parameter [4], allowing applications to choose whether they want the callback mechanism or not. However, ZGC’s reference processor does not take advantage of this distinction, treating all weak references as if they require enqueueing. This results in unnecessary work for references that do not use a `ReferenceQueue`. At the same time, to support this callback mechanism weak semantics are currently implemented through separate `WeakReference` objects. So applications also pay the allocation and processing cost of those objects even if they only need weak reachability.

ZGC’s concurrent architecture makes both of these inefficiencies particularly relevant. Because ZGC processes references concurrently with the application [2, 8], any reduction in per-reference overhead translates into better application performance. This motivates two complementary lines of investigation: optimising the existing `WeakReference` pipeline for the queue-less case, and exploring whether weak semantics expressed through annotated object fields can reduce overhead further by avoiding `WeakReference` objects altogether.

1.2 Problem Statement

The current ZGC reference processing pipeline does not distinguish between weak references with and without a `ReferenceQueue` during its phases [8]. All cleared references are threaded onto a pending list, transferred to the `ReferenceHandler` thread, and iterated regardless of whether they will actually be enqueued or not [5]. Additionally, the intrusive linked-list data structure used for discovered references exhibits poor cache locality [11]. Beyond these pipeline inefficiencies, representing weak semantics through separate `WeakReference` objects introduces object and heap overhead that may be

avoidable if weak reachability is instead expressed directly through annotated object fields.

1.3 Research Questions

The research questions that this thesis aims to address are:

- RQ1** How do modifications to ZGC's processing of `WeakReference` objects without a `ReferenceQueue` affect the wall-clock time of reference processing, the total GC cycle time, the GC memory footprint, and Java heap usage in select workloads?
- RQ2** How does expressing weak reachability through annotated object fields affect reference processing time, GC cycle time, GC memory footprint, and Java heap usage compared to the optimised `WeakReference`-based approach of RQ1?

1.4 Scope and Delimitations

This thesis focuses on the reference processing pipeline within ZGC, specifically targeting weak references without a `ReferenceQueue`. The primary implementation work concerns `WeakReference`-based processing, with weak fields treated as an extension evaluated in the same benchmarking framework. Although the optimisation approaches are described in general terms the implementation and evaluation remain ZGC-specific. Furthermore, some of the optimisations, such as the optimised clear path, may have broader applicability beyond just weak references without a `ReferenceQueue`. However, these broader applications fall outside the scope of this thesis and are not evaluated.

The correctness of the optimisations is only verified through the standard OpenJDK test suite and manual inspection. Performance measurements are conducted on a single hardware platform under controlled conditions and therefore may not generalise to all environments.

2. Background

To answer the research questions posed in Section 1.3, it is necessary to understand key concepts of the Java Virtual Machine’s (JVM) garbage collection mechanisms, the design and implementation of the Z Garbage Collector (ZGC), the Java reference types and their processing pipeline within ZGC, and general principles of garbage collection. This chapter provides an overview of these topics, which will serve as the foundation for the implementation and evaluation of the optimisations explored in this thesis.

2.1 Garbage Collection in the JVM

The JVM provides automatic memory management through garbage collection (GC). Rather than requiring the programmer to explicitly allocate and free memory as in languages such as C or C++, the JVM tracks which objects are reachable from the application’s root set and reclaims the memory of objects that are no longer reachable. The root set consists of all data that is directly accessible to the application, such as local variables on the stack, static fields, and active threads.

The JVM ships with several garbage collector implementations, each designed with different trade-offs between throughput, latency, and memory footprint [12]. The collectors are:

- **Serial GC** A single-threaded collector that performs all garbage collection work using one thread, avoiding inter-thread communication overhead. It is best suited to single-processor systems and small heaps.
- **Parallel GC** A throughput-oriented generational collector that uses multiple GC threads to speed up collection on multiprocessor systems. It is intended for medium to large data sets.
- **G1 GC (Garbage-First)** A mostly concurrent, region-based collector designed to scale from small to large multiprocessor machines. It targets predictable pause times with high probability while maintaining high throughput and is the default collector on most platforms [12].
- **ZGC (Z Garbage Collector)** A scalable, low-latency collector that performs expensive GC work concurrently with application threads. It targets pause times of only a few milliseconds (largely independent of heap size) and supports heap sizes from 8 MB to 16 TB.

This thesis focuses on ZGC, as its low-latency design and concurrent reference processing pipeline can particularly benefit from optimisations that reduce the amount of work done during reference processing.

2.1.1 Garbage Collection Phases

Garbage collection generally occurs multiple times (cycles) during the lifetime of a Java application. Each cycle consists of several distinct phases, each with specific responsibilities. The exact phases and their order can vary between collectors, but a common structure is as follows:

1. **Mark Phase** The collector identifies all live objects by traversing from the root set and marking all reachable objects. Dead objects are those that are not marked. This phase may be performed concurrently with the application or during a stop-the-world pause, depending on the collector design.
2. **Reference Processing Phase** After marking completes, the collector processes *Reference* objects discovered during the mark phase. This phase determines which references have dead referents and prepares them for enqueueing. Like marking, this phase may be concurrent or done during a pause.
3. **Sweep/Reclamation Phase** The collector reclaims memory occupied by dead objects. Some collectors use a sweep algorithm (scanning the heap and freeing unmarked objects) whilst others use evacuation (copying live objects to new memory regions).
4. **Compaction/Relocation Phase** Some collectors compact the heap to eliminate fragmentation, moving live objects to consolidate free space. Others relocate objects incrementally as part of the collection process.

The order and concurrency model of these phases varies significantly between collectors. Serial and Parallel GC typically perform all phases in stop-the-world pauses. G1 performs marking and sweeping mostly concurrently with short pauses for synchronisation. ZGC performs marking, reference processing, and relocation concurrently, using barriers to maintain consistency with concurrent application threads.

2.1.2 Garbage Collection Barriers

In JVM garbage collectors, a *barrier* is a small piece of code inserted around memory accesses so that the collector can maintain correctness while the application is running [13]. The term is slightly misleading: barriers are not global locks and do not stop threads. Instead, they are local checks or bookkeeping actions attached to individual loads and stores of object references [13, 2, 14].

Barriers are especially important in concurrent collectors because GC and application threads observe and modify heap state at the same time. Rather than pausing the application for long periods, the collector uses barriers to spread consistency work across ordinary execution [2, 13].

Two broad classes are common:

- **Load (read) barriers**, executed when a reference is read. These can validate or transform the loaded pointer before it is used by the application [2, 14].
- **Store (write) barriers**, executed when a reference field is overwritten. These record metadata needed by the collector, for example to preserve marking invariants or to track inter-generational pointers [13, 15].

2.1.3 Generational Collection

All JVM garbage collectors (Serial, Parallel, G1, and Z) employ *generational collection*, based on the *weak generational hypothesis*: most objects die young and the ones that don't tend to live for a long time [16]. To employ this, the heap is divided into at least two generations:

- The **young generation**, where newly allocated objects reside. Young generation collections are frequent but fast, since most young objects are short-lived.
- The **old generation**, where objects that have survived multiple young collections are promoted to. Old generation collections are less frequent but more expensive since live objects are more plentiful here.

ZGC adopted generational collection starting with JDK 21 [15]. In generational ZGC, the young and old generations can be collected independently and therefore concurrently. This is relevant to reference processing because references in different generations are handled during different collection cycles.

2.2 Java Reference Types

In addition to ordinary (strong) references, Java provides *reference objects* in the `java.lang.ref` package [3, 5] that allow applications to reference objects without preventing their garbage collection. These reference types are:

1. **SoftReference** The referent field is cleared at the discretion of the GC, typically when memory is low. Commonly used for memory-sensitive caches.
2. **WeakReference** The referent field is cleared as soon as it becomes weakly reachable (i.e., no strong or soft references exist to it). Widely used in caching frameworks, canonical mappings, and listener registrations.
3. **FinalReference** An internal JVM type (not part of the public API) used to implement the `finalize()` mechanism. Objects with finalisers are wrapped in a `FinalReference` so that the GC can schedule finalisation before reclaiming the object. This scheduling is done via a `ReferenceQueue`, which means that `FinalReferences` always have an associated `ReferenceQueue`.

4. **PhantomReference** The referent is never accessible through the reference; `get()` always returns `null`. Used for post-mortem cleanup actions, often to avoid the pitfalls of finalisation. Phantom references must be registered with a `ReferenceQueue` to be useful.

All four types extend the abstract class `Reference<T>` (in the `java.lang.ref` package), which defines the key fields of the reference object and the API for retrieving the referent, clearing the referent field, and checking if it is enqueued.

2.2.1 The Reference Object Lifecycle

Each `Reference` object maintains several fields that track its state [5]:

- **referent field** The referenced object. The GC may clear this field (set it to `null`) when the referent becomes eligible for collection according to the reference's type.
- **queue field** The `ReferenceQueue` this reference is registered with. If the reference was not created with a queue, this field points to the sentinel object `ReferenceQueue.NULL_QUEUE`.
- **next field** Used as a link pointer when the reference is enqueued in a `ReferenceQueue`.
- **discovered field** A field with dual purpose: during GC marking, it serves as a link in the garbage collector's internal *discovered list*; after processing, it is reused as a link in the VM's *pending list* that feeds the `ReferenceHandler` thread. Due to its first use, the GC can detect whether a reference has already been discovered by checking if this field is `null`.

A reference object transitions through the following states during its lifetime [17, 5]:

1. **Active** The reference has been created and the referent is still reachable (or not yet processed).
2. **Pending** The GC has determined the referent is eligible for clearing and has added the reference to the VM's pending list. The `ReferenceHandler` thread will process it.
3. **Enqueued** The `ReferenceHandler` thread has moved the reference into its associated `ReferenceQueue`, where the application can retrieve it via `poll()` or `remove()`.
4. **Inactive** The reference has been dequeued by the application, or was created without a queue and has been cleared. It is no longer tracked by the GC or queuing infrastructure.

An important observation is that references created *without* a `ReferenceQueue` can transition directly from Active to Inactive, bypassing the Pending and Enqueued states entirely. However, as discussed in Section 2.5, the default

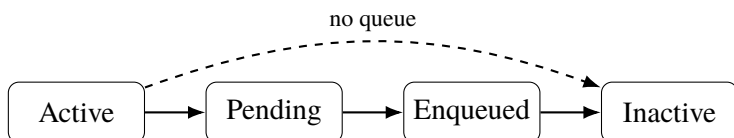


Figure 2.1. Lifecycle of Reference objects. References without a queue can bypass Pending and Enqueued.

implementation does not take advantage of this shortcut. The lifecycle and this direct shortcut path are illustrated in Figure 2.1.

2.2.2 WeakReference and ReferenceQueue

A `WeakReference` provides two constructors (with and without a `ReferenceQueue`), as shown in Listing 2.1 and implemented in OpenJDK [4].

```

1 // Without a queue - queue field set to NULL_QUEUE
2 public WeakReference(T referent) {
3     super(referent);
4 }
5
6 // With a queue
7 public WeakReference(T referent, ReferenceQueue<? super
8     T> q) {
9     super(referent, q);
10 }

```

Listing 2.1: *WeakReference* constructors

When a `WeakReference` is created with a `ReferenceQueue`, the application can be notified when the referent is collected. When created *without* a queue, the reference serves only as a conditional pointer, the application can call `get()` to check whether the referent is still alive, but receives no notification upon clearing.

The `ReferenceQueue` is implemented as a synchronized singly-linked list using the `next` field of the `Reference` objects. It provides `poll()` for non-blocking retrieval and `remove()` for blocking retrieval with an optional timeout [6]. This is the mechanism by which applications can react to the collection of referents, for example to clean up associated resources.

A key insight for this thesis is that Java explicitly supports queue-less weak references through the one-argument `WeakReference` constructor [4]. For workloads that only need liveness probing via `get()`, queue registration is unnecessary. In such cases, the VM still performs enqueue-path preparation work in the default pipeline, which motivated JDK-8029205 and the optimisation work in this thesis [9, 5, 8].

2.3 The JVM Execution Model

To understand how garbage-collector barriers are injected and why ZGC requires different components for each execution tier, we must examine HotSpot’s tiered execution model, which balances startup latency against sustained throughput. Three tiers are relevant to this thesis:

1. **Interpreter.** At JVM startup, an *interpreter* is generated in which each bytecode is mapped to a hand-crafted assembly stub TODO. Methods not yet compiled are executed in this tier. Interpretation is the slowest tier in terms of throughput but has no compilation overhead, and it also gathers profiling data for JIT compilation.
2. **C1 Compiler.** C1 is a fast JIT compiler that performs lightweight optimisations, such as method inlining, constant folding, and null-check elimination TODO. C1 compiles quickly, making it suitable for methods that are called moderately often.
3. **C2 Compiler.** C2 is an aggressive optimising JIT compiler. It consumes profiling data collected by the interpreter and C1 to emit highly optimised native code, applying techniques such as aggressive inlining, loop transformations, and escape analysis TODO. Hot methods are ultimately compiled by C2.

Tiered compilation, the default since TODO, automatically promotes methods through these tiers as their execution frequency increases. A method starts in the interpreter, is promoted to C1-compiled code once it becomes warm, and may eventually be recompiled by C2 once sufficient profiling data has accumulated.

Relevance to GC barriers.

Because a read barrier must intercept every load from the heap, and such loads occur in all three tiers, each tier requires its own barrier implementation. ZGC therefore ships three separate barrier components: `ZBarrierSetAssembler` generates barrier stubs embedded in the interpreter’s dispatch table, `ZBarrierSetC1` inserts barrier nodes during C1 intermediate representation construction, and `ZBarrierSetC2` integrates barriers into the C2 sea-of-nodes graph [14]. All three components obey the same decorator-driven barrier selection logic described in Section 2.4.3.

2.4 The Z Garbage Collector

ZGC is a concurrent, region-based, compacting garbage collector included in the JDK since JDK 11 (experimental) [2] and production-ready since JDK 15 [18]. Starting with JDK 21, ZGC supports generational collection, which is now the default mode [15]. The collector targets millisecond-scale pause times that remain bounded regardless of heap size, and supports heaps ranging from a few hundred megabytes to several terabytes [2]. Achieving

these goals while performing the bulk of GC work concurrently is partly realised by two closely related design decisions: encoding GC state directly in heap pointers, and using barriers to act on that state at every pointer access.

2.4.1 Coloured Pointers and Barriers

ZGC embeds metadata directly in the unused high bits of 64-bit heap pointers. These *colour bits* encode the state of a pointer relative to the current GC phase – for example, whether the pointed-to object has been remapped after relocation, or which generation it belongs to [2, 19]. Because the state is carried in the pointer itself, the GC can determine the status of a reference without loading the object’s header or consulting external tables.

To act on this encoded state, ZGC inserts barriers at every oop load and store. On a load, the barrier reads the colour bits of the retrieved pointer and checks whether they encode a *good* state for the current GC phase. If the pointer is stale – for example because the object has been relocated since the pointer was written – the barrier’s slow path applies the necessary fixup (remapping the pointer to the new address) before returning the address to the caller [2, 14]. Because these fixups happen lazily on access rather than eagerly during relocation, the collector can relocate objects concurrently without stalling the application.

Store barriers serve a complementary role: before overwriting a heap field, the barrier records the field’s previous value in a per-thread store-barrier buffer. This record is consumed by the marking phase to satisfy ZGC’s snapshot-at-the-beginning (SATB) invariant – ensuring that objects that were reachable at the start of a marking cycle are not missed – and by the remembered-set mechanism that tracks cross-generational pointers in the generational case [14].

2.4.2 Concurrent Phases and Parallel Workers

A ZGC collection cycle interleaves short stop-the-world pauses, used for root scanning and phase synchronisation, with longer concurrent phases in which marking, reference processing, and relocation proceed alongside running application threads [2, 8]. The load and store barriers described above are the mechanism that keeps the heap consistent during these concurrent phases: application threads that load a stale pointer automatically fix it up, and stores that would break the marking invariant are recorded for GC consumption.

Within each phase, ZGC distributes its work across multiple GC worker threads. To minimise contention, workers maintain their own thread-local data structures – such as per-worker discovered lists during reference discovery – and merge results into shared state only at well-defined synchronisation points [20, 8]. This design avoids the compare-and-swap contention that would arise from multiple threads updating a single shared resource.

2.4.3 Access Decorators

Throughout the HotSpot codebase, every load or store is annotated with a *decorator set*: a compile-time or runtime bitmask of type `DecoratorSet` (a 64-bit integer) whose set bits describe the semantics of the access [21]. The active `BarrierSet` implementation (e.g. `ZBarrierSet`) inspects this bitmask to decide which barrier variant to invoke, allowing a single generic `HeapAccess<decorators>::load()` call site to resolve to the correct low-level operation at both compile time (for template-specialised paths) and runtime (for dynamically dispatched paths).

The decorator flags are grouped by concern:

- **Reference strength** bits control how the barrier treats reachability. An access with `ON_WEAK_OOP_REF` tells ZGC that the field being read is the referent of a `WeakReference`: the barrier must return null if the referent has been collected, rather than reviving it. The default, when no strength decorator is specified, is `ON_STRONG_OOP_REF`.
- **Access location** bits indicates whether the access targets a heap object field or an off-heap native data structure. Some barrier actions are conditioned on `IN_HEAP`.
- **Barrier strength** bits control how strong barriers should be applied. `AS_RAW` bypasses GC barriers entirely and reads the raw memory value, which is necessary in certain internal GC paths. `AS_NO_KEEPLIVE` performs a correctly remapped access without keeping any objects alive, used in GC-internal scanning paths. `AS_NORMAL` (the default) invokes the full barrier.
- **Memory ordering** bits map to C++ memory-order semantics for the underlying atomic or plain load or store.

For ZGC, when `WeakReference.get()` is executed the referent field load carries `ON_WEAK_OOP_REF`. ZGC's barrier for this decorator checks the colour bits of the loaded pointer and if the pointer is already good (the referent is live and remapped or null), the address is returned directly. If not, the barrier's slow path is invoked. In the slow path, the liveness of the referent is checked and the barrier either returns a coloured null if the referent is no longer live, or a correctly coloured pointer if the referent is live [14, 21].

Store barriers are similarly decorator-driven. A normal store carries the decorator `ON_STRONG_OOP_REF` which triggers ZGC's store barrier, which records the old value in the store-barrier buffer (needed for the snapshot-at-the-beginning marking semantics and for maintaining remembered sets in the generational case).

2.5 Reference Processing Pipeline in ZGC

Reference processing is the mechanism by which the garbage collector handles `Reference` objects discovered during marking. The pipeline has three stages:

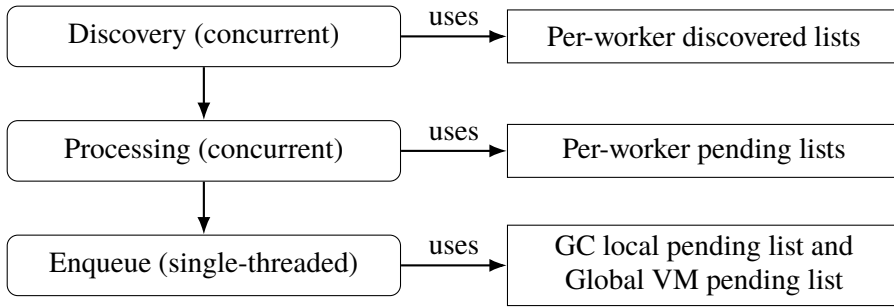


Figure 2.2. ZGC reference-processing pipeline and data flow from discovery to Java-level enqueue handling.

discovery, *processing*, and *enqueueing*. In ZGC, the first two stages run concurrently with the application during the marking phase, while the enqueue stage involves handing off references to the Java-level `ReferenceHandler` thread [8, 5]. An overview of this data flow is shown in Figure 2.2.

2.5.1 Discovery

During the concurrent marking phase, when ZGC encounters a `Reference` object (an instance of `SoftReference`, `WeakReference`, `FinalReference`, or `PhantomReference`), it evaluates whether the reference should be *discovered*, i.e. added to a list for later processing.

The discovery decision depends on several factors:

1. **Is the reference active?** Only active references are eligible. For non-`FinalReference` types, an inactive reference has a null referent field. For `FinalReference`, inactivity is indicated by a non-null next field.
2. **Is the referent still strongly reachable?** If the referent is already marked as strongly live, the reference does not need processing. The garbage collector checks this via its liveness information.
3. **Soft reference policy.** For `SoftReference` objects, the collector applies a policy (based on available memory and the reference’s last access time) to decide whether to keep the referent alive or clear the referent field [10, 22].
4. **Generational considerations.** In ZGC, only references residing in the old generation are discovered; young-generation references are always treated as strong [15, 8]. This is because reference processing introduces additional delay and uncertainty to the duration of the young-generation collection. By treating young references as strong, ZGC can complete young collections more predictably, which decreases the risk of the application running out of memory during a long young collection.

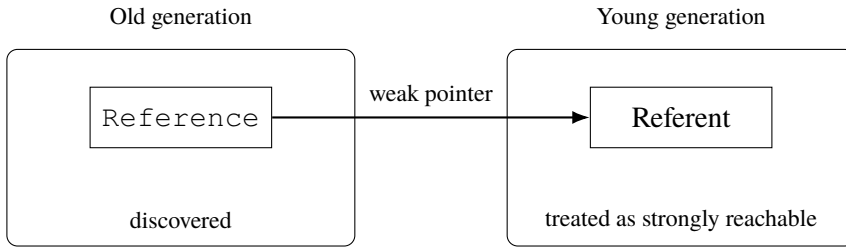


Figure 2.3. If an old-generation `Reference` points to a young referent, the young referent must remain alive during young collection.

Once a reference is selected for discovery, it is recorded on a *discovered* list. In ZGC, each GC worker thread maintains its own discovered-list head, so discovery is implemented as a cheap, non-atomic prepend to a per-worker list [8]. This avoids CAS contention found in shared processors where multiple threads update the same discovered head [10].

2.5.2 Processing

After marking completes, the discovered lists are processed. For each discovered reference, the collector determines the final fate of the referent:

1. If the referent field has become `null` (cleared by the application concurrently), the reference is dropped from the list.
2. If the referent is now strongly reachable (e.g. it was marked as alive after the reference’s discovery), the reference is dropped.
3. If the referent is still unreachable according to the reference’s strength, the referent field is **cleared** (set to `null`), and the reference is prepared for enqueueing.

In ZGC, clearing involves setting the `referent` field to a coloured null pointer via compare-and-swap (CAS) to ensure thread safety since application threads may be concurrently accessing the reference [14, 8].

After clearing, the reference is linked onto the thread-local pending list. This involves writing the reference’s address into the `discovered` field of the previously processed reference. Once the thread has processed all references in its discovered list, the thread-local list is atomically prepended to ZGC’s internal pending list [8].

Generational interactions are important during processing: when a `Refer-ence` object is old but its referent is young, the referent is treated as strongly reachable and the referent field must not be cleared. Processing must therefore verify generation membership and, if appropriate, ensure young-generation liveness is preserved (by marking the young referent), as illustrated in Figure 2.3 [8, 23].

2.5.3 Enqueue

After a reference's referent field has been cleared, the reference must be made available to the application if it was created with a `ReferenceQueue`. This happens through the *pending list* mechanism which feeds the `ReferenceHandler` thread [8, 5]. The handoff process is as follows:

1. The GC has built an internal pending list by linking cleared references together via their `discovered` field.
2. Under the heap lock, the GC atomically prepends its internal pending list onto the global VM pending list.
3. The `ReferenceHandler` daemon thread is notified.
4. The `ReferenceHandler` thread wakes up, atomically retrieves the pending list, and iterates over it. For each reference, it checks whether `queue != NULL_QUEUE`. If the reference has a real queue, it calls `queue.enqueue(this)`; otherwise it simply advances to the next reference.

This handoff reveals an inefficiency: references created without a `ReferenceQueue` are still linked into the pending list, transferred to the `ReferenceHandler` thread, and iterated over only to be skipped [5, 8]. For workloads where the majority of weak references lack queues, this results in a considerable amount of extra work. This behaviour has been reported in an OpenJDK enhancement issue [9]. Avoiding enqueue-path work for references with `NULL_QUEUE` is one of the primary optimisation targets explored in this thesis.

3. Design and Implementation

Three optimisations were implemented to address the inefficiencies in ZGC's processing of `WeakReference` objects without a `ReferenceQueue`. These optimisations are:

- Skipping the enqueue path for weak references without a `ReferenceQueue`, allowing them to be processed without being added to the pending list for the `ReferenceHandler` thread.
- Replacing the intrusive linked list used for discovered references with a dynamic array for weak references without a `ReferenceQueue`.
- Optimising the clear path of references to eliminate unnecessary operations for weak references without a `ReferenceQueue`.

Furthermore, an exploratory implementation of weak fields was implemented to evaluate an alternative approach to expressing weak semantics that avoids the overhead of `WeakReference` objects entirely.

3.1 Skipping Enqueue Path

As described in Section 2.5.2, ZGC enqueues discovered references to the internal pending list regardless of whether they have a `ReferenceQueue` or not. This is inefficient for weak references without a `ReferenceQueue` as discussed in Section 2.5.3. Because of this, the reference processing pipeline was changed so that the enqueue path can be skipped for weak references without a `ReferenceQueue`. This optimisation should reduce the amount of work done by the GC threads during reference processing, resulting in a reduction in reference processing time and GC cycle time.

3.1.1 Implementation of Skipping Enqueue Path

In order to skip enqueueing weak references without a `ReferenceQueue` the `ZReferenceProcessor` class was modified to check if the queue field of a weak reference being discovered is equal to the sentinel value `ReferenceQueue.NULL_QUEUE` before adding it to the discovered list. If the queue field is equal to `ReferenceQueue.NULL_QUEUE` (the weak reference was not created with a queue), it is added to a separate discovered list for weak references without a `ReferenceQueue` instead of being added to the regular discovered list. This separate discovered list is then processed independently

```

1 bool discover_reference(oop ref_obj, ReferenceType type)
2 {
3     zaddress ref = to_zaddress(ref_obj);
4
5     if (!should_discover(ref, type)) {
6         return false;
7     }
8
9     if (type == REF_WEAK && !has_reference_queue(ref)) {
10        weak_no_queue_list_per_worker.append(ref);
11    } else {
12        discovered_list_per_worker.append(ref);
13    }
14
15    mark_discovered(ref);
16    return true;
17 }

```

Listing 3.1: *Pseudocode of the null-queue fast path during discovery. Queue-less weak references are routed to a separate discovered list and processed by a no-enqueue pipeline (see Appendix B.2).*

by the GC threads and none of its weak references are enqueued to the pending list for the `ReferenceHandler` thread. Each GC thread maintains its own separate discovered list for weak references without a `ReferenceQueue` to avoid contention between threads. See Listing 3.1 for a pseudocode summary and Appendix B.2 for the actual implementation.

The reason for separating the discovered lists instead of simply skipping enqueueing for weak references without a `ReferenceQueue` is that two unique reference processing functions can then be used for the two lists: one that includes enqueueing and one that does not. This allows for more efficient processing of the weak references without a `ReferenceQueue`, as it avoids unnecessary checks for whether the weak reference has a queue during processing.

3.2 Dynamic Array for Discovered References

When processing the discovered references, ZGC iterates over the discovered list. This list is, as described in Section 2.5.1, implemented as an intrusive linked list of the discovered references [8]. This iteration over a linked list is inefficient since it has poor cache locality [11]. Because of this, the intrusive linked list was replaced with a dynamic array for weak references without a `ReferenceQueue`. This optimisation should reduce the amount of time spent waiting for the next reference to be loaded during processing thanks

better cache locality [11]. This should reduce reference processing time and GC cycle time thanks to the reduction in waiting time for memory loads during processing.

3.2.1 Implementation of Dynamic Array for Discovered References

The implemented dynamic array, called `ZWeakRefArray`, stores discovered references in a contiguous layout rather than linking them via the `discovered` field. The array is allocated on the C heap and grows dynamically as needed.

Each GC worker thread maintains its own `ZWeakRefArray` instance to avoid contention. During discovery, when a reference should be added to a discovery list, its data is appended to the worker's array instead of being appended to a linked list. The reference's `discovered` field is still set (to a self-referencing pointer) to mark the reference as discovered, preventing duplicate discovery.

During processing, the array is iterated sequentially using index-based access. Because the elements are stored contiguously in memory, this iteration improves cache line utilisation compared to following pointers scattered across the heap [11]. After processing, the array is cleared for the next GC cycle but retains enough capacity to at least hold the same number of references that were processed but not made inactive. This avoids unnecessary reallocations in subsequent cycles when the number of discovered references is relatively stable.

See Listing 3.2 for a pseudocode summary and Appendices B.1 and B.2 for the actual implementation.

3.3 Optimised Clear Path

The standard reference processing path in ZGC uses general-purpose functions designed to handle all reference types uniformly. This generality introduces per-reference overhead that is unnecessary when processing only weak references without a `ReferenceQueue`, where the reference type is always `REF_WEAK` and enqueueing is never required.

The standard function for determining whether a reference's referent should be cleared performs several operations per reference [8, 14]:

1. A load barrier to read the referent from the reference object.
2. A virtual call to determine the reference type (soft, weak, final, or phantom).
3. A compare-and-swap (CAS) operation via a ZGC barrier to atomically set the referent field to a coloured null pointer.

```

1 void discover_reference(oop ref_obj, ReferenceType type)
2 {
3     //...
4     ZWeakRefData data;
5     data.set_from_ref(ref);
6     worker_array.append(data)
7     //...
8 }
9
10 void process_worker_discovered_weak_refs_without_queue (
11     ZWeakRefArray& arr) {
12     size_t dropped = 0;
13     for (size_t i = 0; i < arr.length(); i++) {
14         const ZWeakRefData& data = arr.at(i);
15         data.mark_not_discovered();
16
17         if (try_make_inactive(data)) {
18             cleared_weak_no_queue_count++;
19         } else {
20             dropped++;
21         }
22     }
23     arr.clear_and_reserve(dropped);
24 }

```

Listing 3.2: *Pseudocode for the ZWeakRefArray-backed queue-less weak-reference path. Discovery appends pre-fetched fields and processing scans the array sequentially (see Appendices B.1 and B.2).*

For weak references without a ReferenceQueue, each of these operations can be simplified or eliminated:

1. When combined with the dynamic array optimisation, the referent field address and its value can be pre-loaded during discovery and stored in the array entry. This avoids redundant memory loads during processing. Pre-loading is safe because the only concurrent application-level operations on a Reference's referent are clearing and enqueueing, both of which set the referent field to null rather than to a new non-null value [5, 8].
2. The reference type is known to be REF_WEAK at the call site, so the virtual call to determine the type can be eliminated.
3. The only concurrent modification to the referent field is an application thread setting it to null [5, 14, 8]. Because no thread can set the field to a new non-null value, the CAS operation for clearing the referent can be replaced with a simple store of a coloured null pointer.

These simplifications reduce the instruction count and memory access overhead for each weak reference without a ReferenceQueue during process-

ing, which should lead to a reduction in reference processing time and GC cycle time.

3.3.1 Implementation of Optimised Clear Path

The implementation introduces a new function for determining whether a queue-less weak reference's referent should be cleared (see Appendix B.2). This function takes a data struct (either from the dynamic array or constructed on the fly from the linked list) containing the pre-fetched field addresses and values. This new function has similar semantics to the standard function and the ZGC barriers it uses, but it is specialised for the queue-less weak reference case and can therefore use more efficient operations. See Section 2.4 for more details on ZGC barriers.

The new function checks whether the referent is strongly live via metadata in ZGC's heap management. If the referent is not strongly live and resides in the old generation, its referent field is cleared using a direct pointer store rather than a CAS. This is safe under the constrained update model described above. If the referent is still live (either strongly marked or implicitly by residing in the young generation), the referent field is recoloured with the correct pointer metadata bits, this recolouring, however, has to occur with a CAS so that a null value is not overwritten with a non-null value. Additionally, if the referent is young and the young generation is currently in its mark phase, the referent is marked to preserve cross-generational liveness [8, 14, 23].

The `discovered` field is also cleared using a direct store to the field address rather than through an atomic access function, since no other thread modifies this field after discovery in this processing path [8].

See Listing 3.3 for a pseudocode summary and Appendix B.2 for the actual implementation.

3.4 Weak Fields

The three core optimisations above operate within the existing `WeakReference` object model. However, this model has an inherent structural overhead: each weak relation requires a separate Java object with queue and list-link metadata, and the GC must still discover and process that object. To answer Research Question 2, an exploratory weak-field mechanism was implemented to test whether expressing weak reachability directly on object fields can reduce this overhead. This mechanism allows developers to annotate object fields as weak, indicating that the field should be treated with weak semantics by the GC. The GC then applies weak reachability rules directly to the annotated field without requiring a separate `WeakReference` object. This should reduce reference processing time slightly and reduce Java heap usage by eliminating the need for `WeakReference` objects. The reduction in Java heap usage

should also lead to a reduction in GC cycle time, since the cost of garbage collection is proportional to the number of live objects [13].

3.4.1 Implementation of Weak Fields

The implementation includes changes in three layers of the JVM: a Java-level annotation, GC-side field discovery and processing, and compiler/interpreter support for the annotation metadata.

Annotation

A new runtime-retained field annotation, `@java.lang.ref.weak`, was added to the `java.lang.ref` package. The annotation targets only fields and is retained at runtime so that the VM can inspect it during class loading. See Listing 3.4 for the annotation definition. During class-file parsing, the VM recognises the `@weak` annotation and sets a `weak` flag in the field's metadata.

Discovery

Weak-field discovery occurs during ZGC's old-generation marking phase. As the marking closure iterates over each object's reference fields, it checks whether the current field carries the `weak` flag in its metadata. If the field is annotated as weak, the marking closure diverts it to the weak-field discovery path instead of applying the normal strong mark barrier [8].

The discovery function first checks whether the field should be discovered at all. Fields that already point to null or to a strongly live object (identified by their ZGC pointer metadata bits) are skipped, since no clearing action will be needed for them. Fields in young-generation objects are also skipped, because young-generation weak-field processing is not supported by this implementation. If the field passes these checks, its address and current value are recorded as a `ZWeakFieldData` struct and appended to the GC worker thread's per-worker `ZWeakFieldArray`. See Listing 3.5 for a pseudocode summary and Appendix B.3.3 for the actual implementation.

Each GC worker thread maintains its own `ZWeakFieldArray` instance to avoid contention. The array is structurally identical to the `ZWeakRefArray` used for queue-less weak references: it stores entries contiguously on the C heap and grows dynamically (see Appendix B.3.2).

Processing

During the reference processing phase, each GC worker iterates over its `ZWeakFieldArray` and applies a clean barrier to each entry. The clean barrier uses the data that was recorded during discovery. If the referent is no longer strongly live, the barrier clears the field by storing a coloured null pointer; otherwise it recolours the pointer with the correct metadata bits. This is the same barrier used by the standard `WeakReference` processing

path [14]. After processing, the array is cleared but retains enough capacity to hold the entries that were not cleared, following the same capacity-retention strategy as `ZWeakRefArray`. See Listing 3.6 for a pseudocode summary and Appendix B.3.3 for the actual implementation.

Compiler and Interpreter Support

The `@weak` annotation metadata is also made available to the JIT compilers and interpreter. This is needed to ensure that any loads or stores to a weak field outside of GC processing are handled correctly with respect to the weak semantics. This is done by adding a decorator to loads or stores that are done to a field that carries the `weak` flag in its metadata. ZGC then inserts barriers that apply the necessary semantics on all loads and stores with this decorator. Without the load decorator, a load from a weak field could return a non-null value that has already been cleared by the GC, which would quickly lead to a crash. The store decorator is needed so that the GC doesn't need to record the previous value of the field (for preserving the snapshot-at-the-beginning mark semantics), as needed in the default case. See Section 2.4.3 for more details on the decorators and Appendix B.3.5 for the implementations.


```

1 bool try_make_inactive_fast(const ZWeakRefData& data) {
2     // if used with the dynamic array
3     zaddress referent = data.referent_addr;
4
5     // if used with the linked list
6     zaddress referent = load_barrier(data.
7         referent_field_addr);
8
9     // Young referents are implicitly kept alive by
10    generational rules.
11    if (is_young(referent)) {
12        if (young_mark_is_active()) {
13            mark_young_root(referent);
14        }
15        return false; // keep as active
16    }
17
18    // Old referent: clear only if not strongly live.
19    if (!is_strongly_live(referent)) {
20        *data.referent_field_addr = to_zpointer(zaddress
21            ::null);
22        return true; // became inactive
23    }
24
25    // Still live: recolour pointer metadata, but do not
26    // overwrite null.
27    zpointer observed = data.referent_field_value;
28    zpointer recoloured = remap_with_good_metadata(
29        referent);
30    cas_referent_if_non_null(data.referent_field_addr,
31        observed, recoloured);
32    return false;
33 }

```

Listing 3.3: Pseudocode for the specialised queue-less weak-reference inactive check. The old-generation clear path uses direct stores, while recolouring uses CAS to avoid null-to-non-null races (see Appendix B.2).

```

1 package java.lang.ref;
2
3 @Retention(RetentionPolicy.RUNTIME)
4 @Target(ElementType.FIELD)
5 public @interface weak {}

```

Listing 3.4: The `@weak` annotation. Applied to a field, it instructs the GC to treat the field value with weak-reference semantics (see Appendix B.3.1).

```

1 // During marking: the closure checks each reference
  field
2 void do_oop(oop* p) {
3     if (is_weak_annotated_field(p)) {
4         if (discover_weak_field(p)) {
5             return; // discovered, skip strong marking
6         }
7     }
8     // ... normal strong marking ...
9 }
10
11 bool discover_weak_field(oop* field) {
12     encountered_weak_fields_count++;
13
14     if (is_young(field)) {
15         return false; // young generation not supported
16     }
17
18     if (is_null(field) || is_strongly_live(field)) {
19         return false; // nothing to clear
20     }
21
22     discovered_weak_fields_count++;
23     ZWeakFieldData data(field);
24     worker_weak_field_array.append(data);
25     return true;
26 }

```

Listing 3.5: Pseudocode for weak-field discovery during marking. Annotated fields are diverted from the strong mark barrier and recorded in a per-worker array (see Appendix B.3.3).

```

1 void process_worker_discovered_weak_fields(
2     ZWeakFieldArray& weak_fields) {
3     size_t dropped = 0;
4     for (size_t i = 0; i < weak_fields.length(); i++) {
5         const ZWeakFieldData& data = weak_fields.at(i);
6         if (clean_barrier_weak(data)) {
7             cleared_weak_fields_count++;
8         } else {
9             dropped++;
10        }
11    }
12    weak_fields.clear_and_reserve(dropped);
13 }

```

Listing 3.6: *Pseudocode for weak-field processing. Each entry is passed to the clean barrier, which clears dead referents and recolours live ones (see Appendix B.3.3).*

4. Evaluation

The implemented optimisations described in Chapter 3 were evaluated through custom microbenchmarks designed to synthetically target the reference processing pipeline. The benchmarks were executed on a supercomputer cluster under controlled conditions, and the resulting GC timing statistics and memory usage metrics were collected to enable comparison between the baseline and optimised variants.

4.1 Optimisation Variants

Nine optimisation variants were evaluated in total. Eight of them cover all combinations of the three `WeakReference` pipeline optimisations described in Chapter 3: the optimised clear path, skipping enqueue for queue-less references via separate discovered lists, and the dynamic array. Table 4.1 shows which optimisations each variant enables. The `none` variant serves as the unmodified baseline. The remaining variants enable a subset of the three optimisations, allowing their individual and combined effects to be isolated. The `all` variant enables all three simultaneously. The ninth and final variant, `weak_fields`, corresponds to the `@weak` field annotation described in Section 3.4.

Each variant is implemented as a set of patches under `patches/<variant>/. All variants except for dyn_only and weak_fields also include a shared base patch with infrastructure changes common to those optimisations. Table 4.2 lists the resulting diff sizes; full per-file breakdowns are available in Appendix C. A build script (scripts/build_configs.sh) automates applying the appropriate patches and building each JDK binary.`

4.2 Performance Evaluation

To evaluate the performance impact of the optimisation approaches, four microbenchmarks were used under controlled conditions: two `WeakReference`-based benchmarks and two weak-field counterparts with analogous workload structure. The benchmarks and their execution environment are described below.

Table 4.1. *Optimisation variants and their enabled optimisations. A tick indicates the optimisation is active; a dash indicates it is disabled.*

Variant	Optimised Clear Path	Skip Enqueue	Dynamic Array
none	—	—	—
dyn_only	—	—	✓
sep_only	—	✓	—
sep_dyn	—	✓	✓
clear_path_only	✓	—	—
clear_path_dyn	✓	—	✓
clear_path_sep	✓	✓	—
all	✓	✓	✓
weak_fields	—	—	—

Table 4.2. *Files changed and diff size per variant relative to upstream. Variants marked with † include the shared base patch (8 files, +43/−16 lines) in their totals.*

Variant	Modified	New	Ins.	Del.	Net
all†	10	1	+397	−68	+329
clear_path_dyn†	10	1	+399	−84	+315
clear_path_only†	10	0	+188	−41	+147
clear_path_sep†	10	0	+214	−46	+168
dyn_only	2	1	+205	−69	+136
sep_dyn†	10	1	+360	−61	+299
sep_only†	10	0	+173	−43	+130
weak_fields	28	2	+748	−232	+516

4.2.1 Benchmark Design

Two benchmark families were used to evaluate the optimisation approaches. The first family isolates reference-processing cost around a single shared referent, and the second is a more realistic multi-object workload that introduces greater allocation and liveness variability. Each family was implemented both with `WeakReference` objects and with weak fields. In order to enable `WeakReference`-to-weak-field comparison, both benchmark families use holder objects that either contain a `WeakReference` or an annotated weak field.

Single-Object Benchmark

The Single-Object benchmarks allocate 10 million holder objects with either a `WeakReference` object or an annotated weak field each that all point to a single target object. The holder objects are allocated and stored in a randomised order using a Fisher-Yates shuffle [24] to avoid artificial locality patterns. None

Table 4.3. *Cluster node hardware specification*

Property	Value
Total nodes	80
CPU	AMD EPYC 9454P (Zen 4), 2.75 GHz
Physical cores	48
Threads	96
Memory	768 GiB
Scratch	1.7 TiB
GPUs	None

of the weak references are created with a `ReferenceQueue`. After a brief holding period, the single strong reference to the target object is released and a GC cycle is triggered manually. Because all weak references become clearable simultaneously in a single GC cycle, this benchmark maximises the per-cycle reference processing load and isolates the cost of the processing pipeline itself.

Multi-Object Benchmark

The multi-object benchmarks represent a slightly more realistic workload. They allocate approximately 7.3 million objects, each containing a randomly sized byte array payload (between 509 and 1097 bytes). Each object has a strong reference and a corresponding holder object with either a `WeakReference` object or an annotated weak field. The allocation and storing order of both strong references and holder objects are randomised using separate Fisher-Yates shuffles [24] to avoid artificial locality patterns.

Unlike the Single-Object benchmark, the strong references are not all released at once. Instead, the benchmark proceeds through five rounds of clearing. In each round, 20% of the strong references are nulled in a randomised order, making the corresponding objects eligible for garbage collection. Between rounds, the benchmark sleeps to allow GC cycles to discover and process the weak references whose referent now is unreachable. This gradual release pattern means that weak references/fields are discovered and processed across multiple GC cycles rather than in a single burst.

4.2.2 Execution Environment

The benchmarks were executed on UPPMAX, a supercomputer cluster at Uppsala University. Table 4.3 lists the hardware specification of the cluster nodes. Each variant was run using its pre-built JDK binary (see Section 4.1). Table 4.4 lists the JVM options used for each benchmark family.

A 10-second cooldown period was inserted between variant runs to allow the system to return to a stable state.

4.2.3 Measurement Methodology

Each benchmark was run for a total of 100 iterations. Jobs were submitted via SLURM and allocated two nodes running in exclusive mode. Hyperthreading was disabled at the SLURM level, so only the 48 physical cores per node were available. To parallelise execution across the two nodes, iterations were split across 8 parallel instances (4 per node). Each instance was pinned via `taskset` to a disjoint set of 10 JVM cores and 2 auxiliary cores (12 cores per instance), preventing interference between concurrently running instances. Each instance ran its share of the 100 iterations preceded by 1 warm-up iteration whose results were discarded. The warm-up executed the full benchmark workload for every variant. Within each measurement iteration, all nine optimisation variants were run in sequence with a 10-second cooldown between each. This interleaved ordering mitigates systematic drift in system conditions affecting one configuration more than others.

During each run, a memory monitoring script samples memory metrics from the process at high frequency. See Table 4.5 for the memory metrics collected.

In addition to the continuous memory samples, GC statistics are parsed from the log files of each run. For each GC metric, the parser extracts the Total Avg/Max pair and unit reported by ZGC. The single-object benchmark concentrates its entire weak-reference load into a single collection cycle, so the per-run *maximum* is the relevant statistic. But since the multi-object benchmark spreads work across many cycles the per-run *average* is used instead. Table 4.6 lists the GC log metrics collected.

4.3 Presentation of Results

Results are presented for all nine variants listed in Table 4.1: the unmodified baseline (`none`), the seven `WeakReference` pipeline variants, and `weak_fields`. This enables two complementary comparisons:

- **Intra-model comparison** among the `WeakReference` variants, isolating the effect of each optimisation and their interactions relative to the baseline.
- **Inter-model comparison** between the best-performing `WeakReference` variants and the weak-field representation, evaluating whether avoiding `WeakReference` objects altogether offers additional benefit.

Two families of plots are used. GC timing behaviour is shown as box plots (Figures 4.1 to 4.3 for the single-object benchmark; Figures 4.4 to 4.6 for the multi-object benchmark). Each box spans the interquartile range (IQR) from the 25th to the 75th percentile, with a horizontal line at the median. Whiskers extend to the most extreme data point within $1.5 \times \text{IQR}$ of the box edge, and points beyond the whiskers are plotted individually as outliers. For the single-object benchmark, each data point is the per-run maximum of the metric; for

the multi-object benchmark it is the per-run average, following the rationale in Section 4.2.3.

Memory behaviour is shown as continuous time-series traces (Figure 4.7 for single-object; Figure 4.8 for multi-object). Each trace plots the mean value across all runs as a solid line, with a shaded band showing ± 1 standard deviation. Three metrics are tracked: total process RSS, GC auxiliary memory, and Java heap usage.

Table 4.4. *JVM options per benchmark family*

Option	Benchmark	Purpose
<code>-Xms7g -Xmx7g</code>	Single	Fixed 7 GB heap
<code>-Xms8g -Xmx8g</code>	Multi	Fixed 8 GB heap
<code>-XX:+UseZGC</code>	Both	Enable ZGC
<code>-XX:InitialTenuringThreshold=1</code>	Both	Promote objects to old generation after one cycle
<code>-XX:MaxTenuringThreshold=1</code>	Both	Cap tenuring at 1
<code>-Xlog:gc+stats,gc+ref</code>	Both	GC statistics and reference processing logs
<code>-XX:NativeMemoryTracking=summary</code>	Both	Native Memory Tracking
<code>-XX:ZCollectionIntervalMajor=0.5</code>	Multi	Force major collection every 0.5 s
<code>-XX:+ZCollectionIntervalOnly</code>	Multi	Only timer-triggered major collections

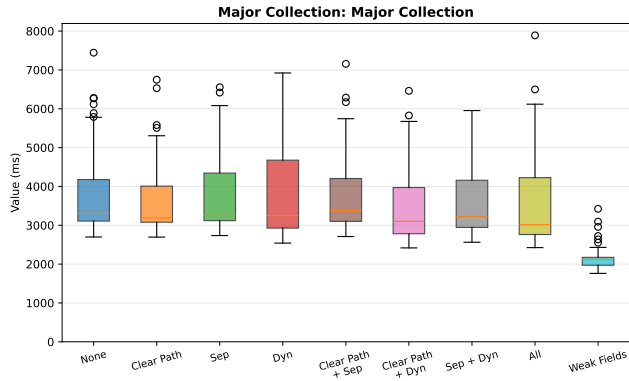
Table 4.5. *Continuous monitoring metrics used in the measurement pipeline.*

Metric	Units	Description
Timestamp	ms	Sample timestamp used to align and aggregate runs over time.
RSS	kB	Process resident set size (RSS), representing total physical memory in use by the JVM process.
GC Committed Memory	kB	GC-related memory currently committed by the JVM.
Java Heap	kB	Java heap usage.

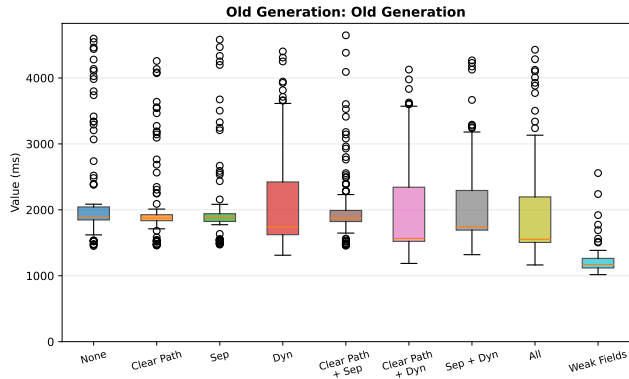
Table 4.6. *GC log metrics used in the measurement pipeline. For the single-object benchmark, per-run maximum values are used; for the multi-object benchmark, per-run averages are used.*

Metric	Units	Description
Major Collection	ms	End-to-end duration of major collections.
Old Generation	ms	Total time spent in old-generation collection work.
Old Phase: Concurrent Mark	ms	Time spent in old-generation concurrent marking.
Old Phase: Concurrent Process Non-Strong	ms	Time spent processing non-strong references in the old generation.
Old Phase: Concurrent Relocate	ms	Time spent in old-generation concurrent relocation.
Young Generation [†]	ms	Total time in young-generation collection. For multi-object, zero values (cycles without young-gen work) are excluded.
Young Phase: Concurrent Mark	ms	Time spent in young-generation concurrent marking.
Young Phase: Concurrent Relocate	ms	Time spent in young-generation concurrent relocation.

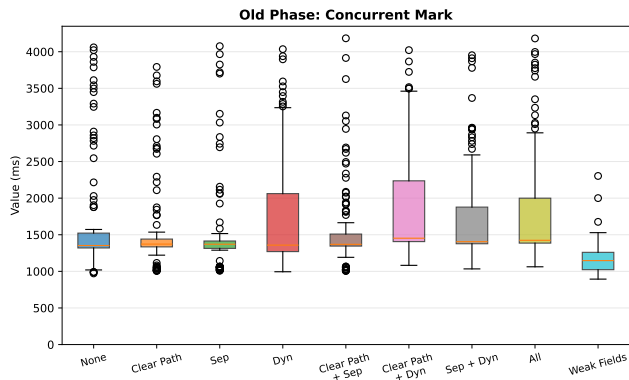
[†]Multi-object uses “Young Generation: Young Generation”; single-object uses “Young Generation: Young Generation (Promote All)”.



(a) Major Collection

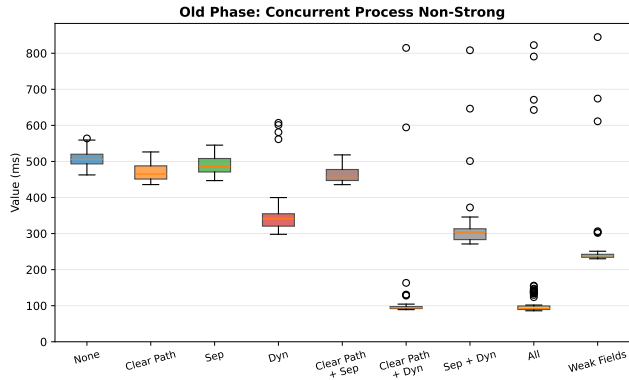


(b) Old Generation

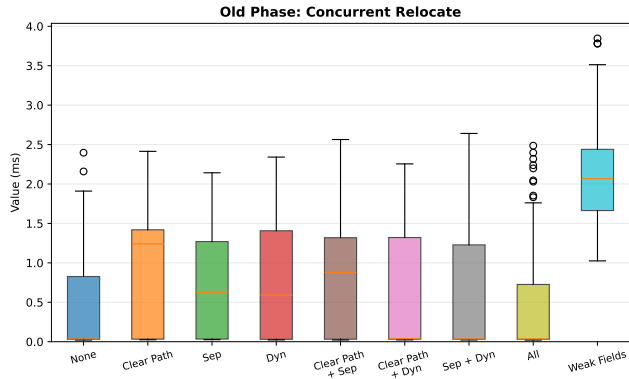


(c) Old Phase: Concurrent Mark

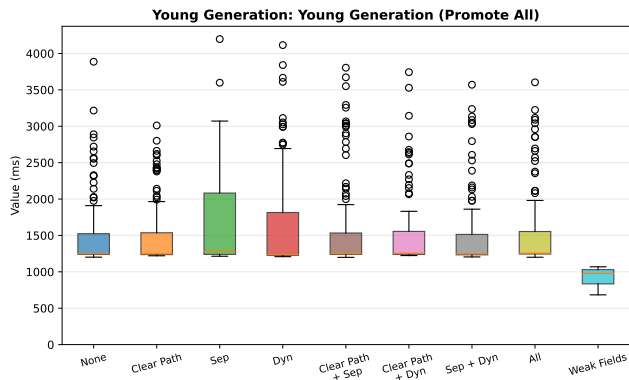
Figure 4.1. Box plots of GC timing metrics across evaluated configurations for the single-object benchmark (per-run maxima), part 1.



(a) Old Phase: Concurrent Process Non-Strong

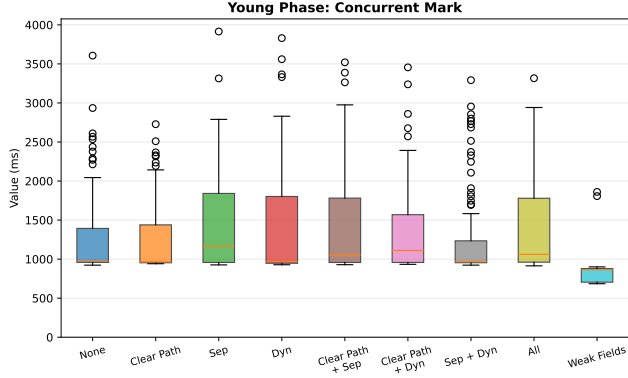


(b) Old Phase: Concurrent Relocate

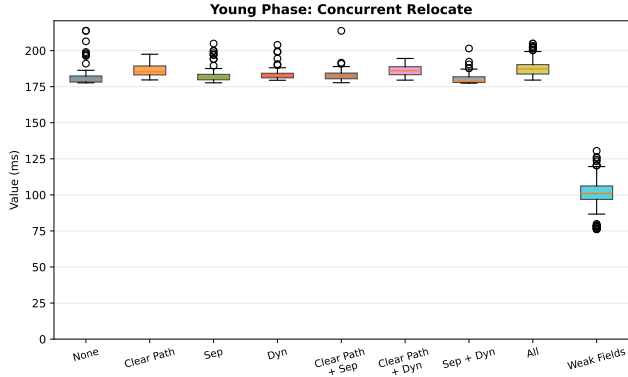


(c) Young Generation (Promote All)

Figure 4.2. Box plots of GC timing metrics across evaluated configurations for the single-object benchmark (per-run maxima), part 2.

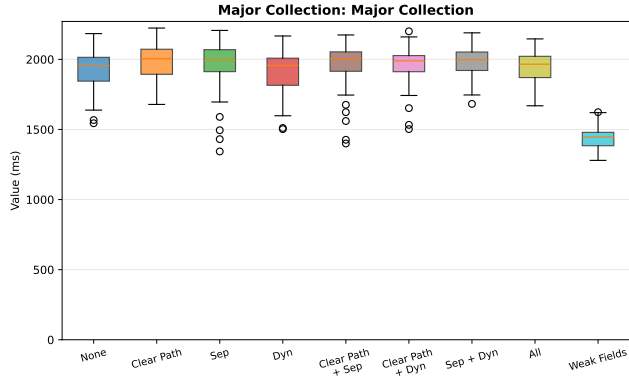


(a) Young Phase: Concurrent Mark

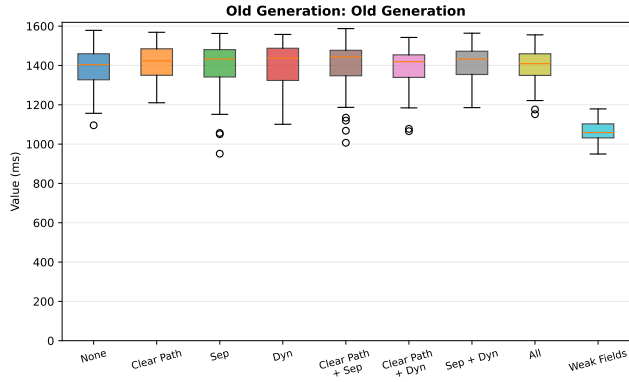


(b) Young Phase: Concurrent Relocate

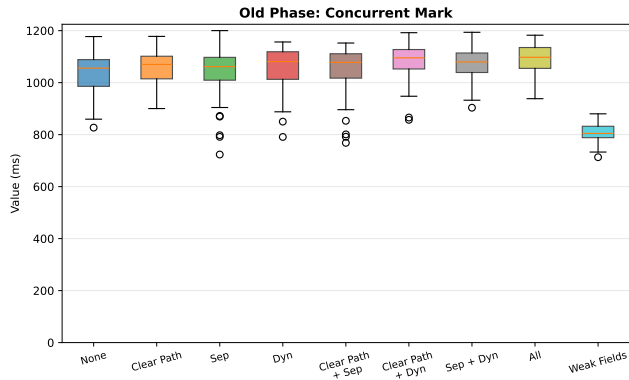
Figure 4.3. Box plots of GC timing metrics across evaluated configurations for the single-object benchmark (per-run maxima), part 3.



(a) Major Collection

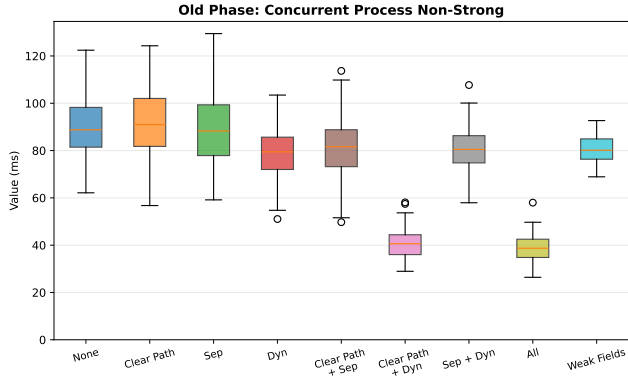


(b) Old Generation

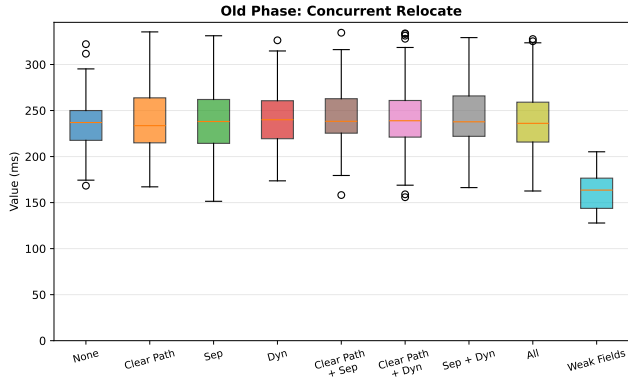


(c) Old Phase: Concurrent Mark

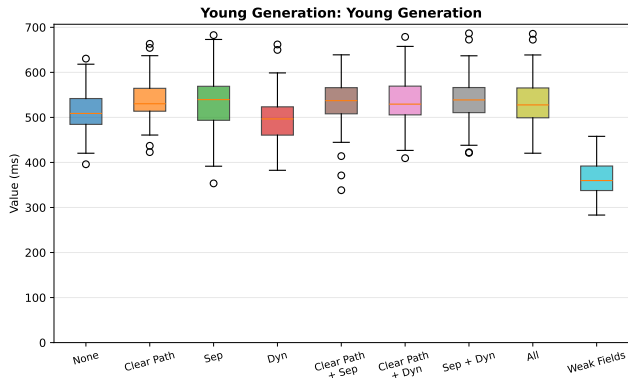
Figure 4.4. Box plots of GC timing metrics across evaluated configurations for the multi-object benchmark (per-run averages), part 1.



(a) Old Phase: Concurrent Process Non-Strong

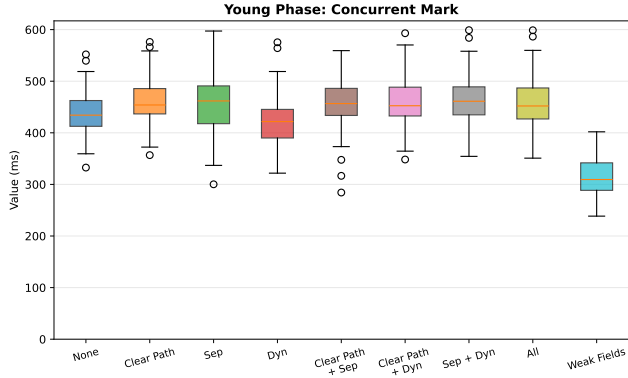


(b) Old Phase: Concurrent Relocate

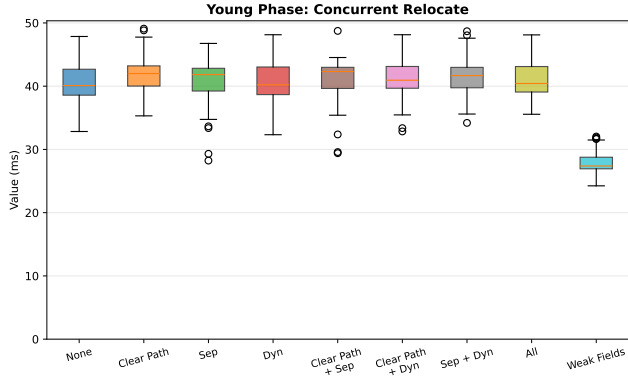


(c) Young Generation

Figure 4.5. Box plots of GC timing metrics across evaluated configurations for the multi-object benchmark (per-run averages), part 2.

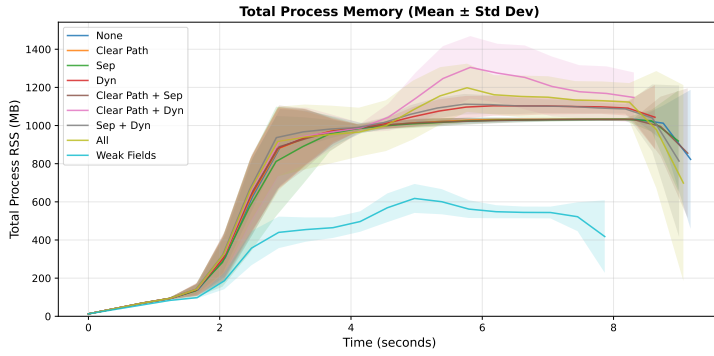


(a) Young Phase: Concurrent Mark

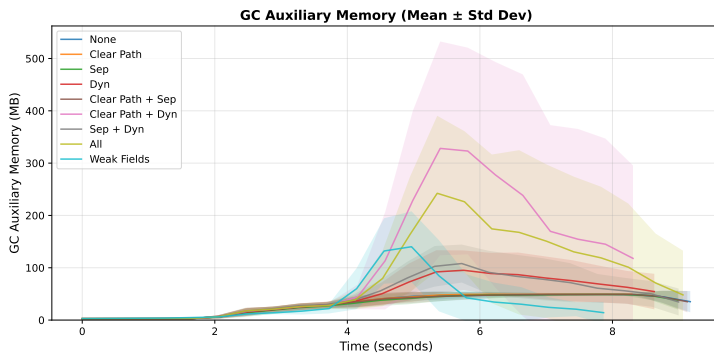


(b) Young Phase: Concurrent Relocate

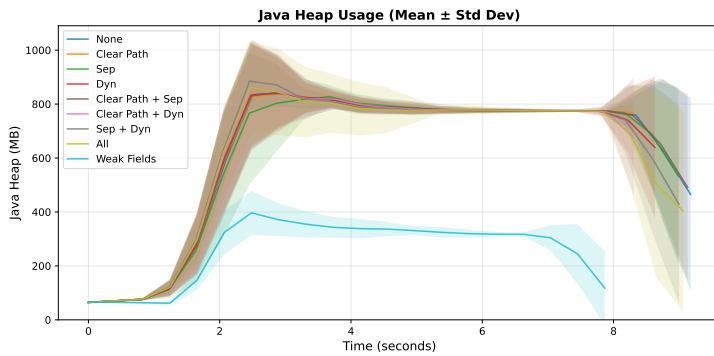
Figure 4.6. Box plots of GC timing metrics across evaluated configurations for the multi-object benchmark (per-run averages), part 3.



(a) Total Process RSS

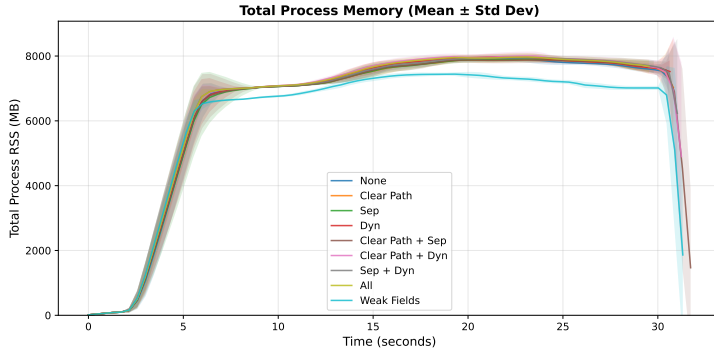


(b) GC Auxiliary Memory

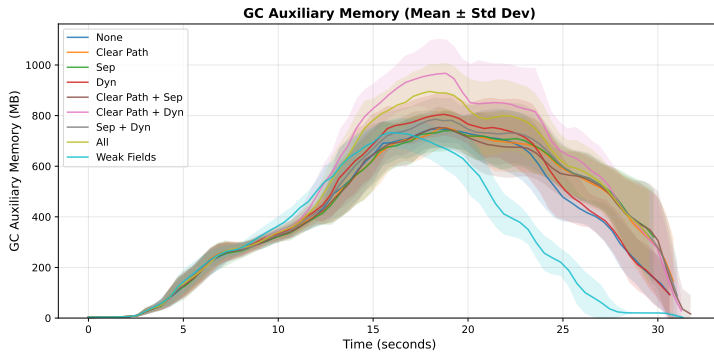


(c) Java Heap

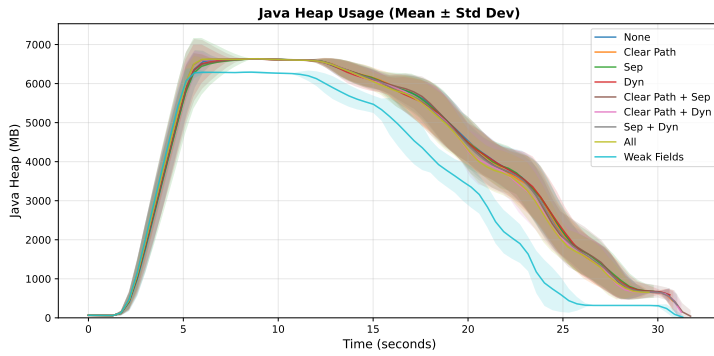
Figure 4.7. Continuous memory traces for the single-object benchmark (mean $\pm$ 1 std dev across runs).



(a) Total Process RSS



(b) GC Auxiliary Memory



(c) Java Heap

Figure 4.8. Continuous memory traces for the multi-object benchmark (mean $\pm$ 1 std dev across runs).

5. Analysis of Results

6. Related Work

7. Discussion

7.1 Future Work

Can the optimisations developed for ZGC be applied to other garbage collectors in the JVM, such as the G1 garbage collector, and if so, what performance improvements can be observed?

8. Conclusion

References

- [1] Wikipedia, “Programming languages used in most popular websites.” https://en.wikipedia.org/wiki/Programming_languages_used_in_most_popular_websites, 2025. Accessed: 2026-03-12.
- [2] OpenJDK, “Jep 333: Zgc: A scalable low-latency garbage collector (experimental).” <https://openjdk.org/jeps/333>, 2018. Accessed: 2026-03-12.
- [3] Oracle, “java.lang.ref (java se 25).” <https://docs.oracle.com/en/java/javase/25/docs/api/java.base/java/lang/ref/package-summary.html>, 2025. Accessed: 2026-03-12.
- [4] OpenJDK, “Openjdk source: java.lang.ref.weakreference.java.” <https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/lang/ref/WeakReference.java>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [5] OpenJDK, “Openjdk source: java.lang.ref.reference.java.” <https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/lang/ref/Reference.java>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [6] OpenJDK, “Openjdk source: java.lang.ref.referencequeue.java.” <https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/lang/ref/ReferenceQueue.java>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [7] OpenJDK, “Openjdk source: java.util.weakhashmap.java.” <https://github.com/openjdk/jdk/blob/master/src/java.base/share/classes/java/util/WeakHashMap.java>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [8] OpenJDK, “Openjdk source: Zgc reference processor (zreferenceprocessor.cpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/z/zReferenceProcessor.cpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [9] OpenJDK Bug System, “Unnecessary pending-list traversal for references without referencequeue.” <https://bugs.openjdk.org/browse/JDK-8029205>, n.d. Accessed: 2026-02-24.
- [10] OpenJDK, “Openjdk source: Shared reference processor (referenceprocessor.cpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/shared/referenceProcessor.cpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [11] U. Drepper, “What every programmer should know about memory,” tech. rep., Red Hat, Inc., 2007. Accessed: 2026-04-07.
- [12] Oracle, “Available collectors.” <https://docs.oracle.com/en/java/javase/25/gctuning/available-collectors.html>, 2025. Java Platform, Standard Edition 25 Garbage Collection Tuning Guide. Accessed: 2026-02-24.
- [13] Oracle, “Garbage collector implementation.” <https://docs.oracle.com/en/java/javase/25/gctuning/garbage-collector-implementation.html>, 2025. Java Platform, Standard Edition 25 Garbage Collection Tuning Guide. Accessed: 2026-04-07.

- [14] OpenJDK, “Openjdk source: Zgc barrier implementation (zbarrier.cpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/z/zBarrier.cpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [15] OpenJDK, “Jep 439: Generational zgc.” <https://openjdk.org/jeps/439>, 2023. Accessed: 2026-03-12.
- [16] H. Lieberman and C. Hewitt, “A real time garbage collector based on the lifetimes of objects,” Tech. Rep. AIM-569a, MIT Artificial Intelligence Laboratory, Oct. 1981. Accessed: 2026-02-24.
- [17] Oracle, “Reference (java se 25).” <https://docs.oracle.com/en/java/javase/25/docs/api/java.base/java/lang/ref/Reference.html>, 2025. Accessed: 2026-03-12.
- [18] OpenJDK, “Jep 377: Zgc: A scalable low-latency garbage collector (production).” <https://openjdk.org/jeps/377>, 2020. Accessed: 2026-03-12.
- [19] OpenJDK, “Openjdk source: Zgc address representation (zaddress.hpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/z/zAddress.hpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [20] OpenJDK, “Openjdk source: Zgc worker infrastructure (zworkers.cpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/z/zWorkers.cpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [21] OpenJDK, “Openjdk source: Access decorators (accessdecorators.hpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/oops/accessDecorators.hpp>, 2026. GitHub code blob. Accessed: 2026-04-16.
- [22] OpenJDK, “Openjdk source: Reference policy (referencepolicy.cpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/shared/referencePolicy.cpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [23] OpenJDK, “Openjdk source: Generational zgc implementation (zgeneration.cpp).” <https://github.com/openjdk/jdk/blob/master/src/hotspot/share/gc/z/zGeneration.cpp>, 2026. GitHub code blob. Accessed: 2026-03-24.
- [24] D. E. Knuth, *The Art of Computer Programming, Volume 2: Seminumerical Algorithms*. Addison-Wesley, 3rd ed., 1997. Section 3.4.2, Algorithm P.

Appendix A.

Usage of Generative Artificial Intelligence Tools

The generative artificial intelligence (GenAI) tool GitHub Copilot was used as support during this thesis. It was used under a student licence via Uppsala University. The tool was used to accelerate practical work and improve writing quality, while all final technical decisions, implementation choices, and evaluations remained under the author's control.

A.1 Use During Implementation

GenAI was used as support during coding of the implementation, primarily for discussing alternative implementation approaches, suggesting code edits through auto-complete, and implementing refactorings of already functioning code.

A.2 Use for Tests and Scripts

GenAI was used to completely generate several test files and automation scripts used in this project, including scripts for benchmarking workflows and result-processing utilities. These generated artefacts were reviewed and validated through repeated execution.

A.3 Use During Writing

GenAI was used as writing support in three main ways:

- Suggesting rephrasing for clarity, tone, and structure.
- Suggesting and correcting LaTeX formatting and fixing compilation or syntax errors.
- Generating first drafts of information-dense sections, especially where the text was based directly on OpenJDK source code and implementation details.

All GenAI-generated text was manually edited and verified against primary sources before inclusion in the final manuscript.

Appendix B.

Source Code

This appendix provides the key source code for the implemented optimisations. The code for the three core optimisations is from the combined (all) configuration variant, which includes all three optimisations. The code for the weak-field mechanism is from the weak_fields configuration variant. Complete sources for all variants are available in the patches/ directory of the project repository.

B.1 Dynamic-array Data Structure

File: zWeakRefArray.hpp

```
1  /*
2   * Copyright (c) 2025, Oracle and/or its affiliates. All rights
   * reserved.
3   * DO NOT ALTER OR REMOVE COPYRIGHT NOTICES OR THIS FILE HEADER.
4   *
5   * This code is free software; you can redistribute it and/or
   * modify it
6   * under the terms of the GNU General Public License version 2
   * only, as
7   * published by the Free Software Foundation.
8   *
9   * This code is distributed in the hope that it will be useful,
   * but WITHOUT
10  * ANY WARRANTY; without even the implied warranty of
   * MERCHANTABILITY or
11  * FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public
   * License
12  * version 2 for more details (a copy is included in the
   * LICENSE file that
13  * accompanied this code).
14  *
15  * You should have received a copy of the GNU General Public
   * License version
16  * 2 along with this work; if not, write to the Free Software
   * Foundation,
17  * Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA.
```

```

18  *
19  * Please contact Oracle, 500 Oracle Parkway, Redwood Shores,
    CA 94065 USA
20  * or visit www.oracle.com if you need additional information
    or have any
21  * questions.
22  */
23
24 #ifndef SHARE_GC_Z_ZWEAKREFARRAY_HPP
25 #define SHARE_GC_Z_ZWEAKREFARRAY_HPP
26
27 #include "gc/z/zAddress.hpp"
28 #include "memory/allocation.hpp"
29 #include "utilities/debug.hpp"
30 #include "utilities/powerOfTwo.hpp"
31 #include "utilities/globalDefinitions.hpp"
32 #include <string.h>
33
34 struct ZWeakRefData {
35     zpointer* referent_field_addr;
36     zaddress* discovered_field_addr;
37     zaddress referent_addr;
38     zpointer referent_field_value;
39 };
40
41
42 class ZWeakRefArray : public AnyObj {
43 private:
44     ZWeakRefData* _data;
45     const size_t _ZWeakRefData_size = sizeof(ZWeakRefData);
46     size_t _length;
47     size_t _capacity;
48
49     void expand_to(size_t new_capacity) {
50         assert(new_capacity > _capacity, "expected growth but %ld <=
            %ld", new_capacity, _capacity);
51
52         ZWeakRefData* new_data = (ZWeakRefData*)AllocateHeap(
            new_capacity * _ZWeakRefData_size, mtGC);
53
54         if (_length > 0) {
55             // Use memcpy for trivially copyable types - much faster
            // than element-by-element copy
56             memcpy(new_data, _data, _length * _ZWeakRefData_size);
57         }
58
59         if (_data != nullptr) {
60             FreeHeap(_data);
61         }
62
63         _data = new_data;

```

```

64     _capacity = new_capacity;
65 }
66
67 void grow(size_t min_capacity) {
68     size_t new_capacity = MAX2((size_t) 8, next_power_of_2(
69         min_capacity - 1));
70     expand_to(new_capacity);
71 }
72
73 public:
74     ZWeakRefArray() : _data(nullptr), _length(0), _capacity(0) {}
75
76     ~ZWeakRefArray() {
77         if (_data != nullptr) {
78             FreeHeap(_data);
79             _data = nullptr;
80         }
81
82     void append(const ZWeakRefData& entry) {
83         if (_length >= _capacity) {
84             grow(_length + 1);
85         }
86         _data[_length] = entry;
87         _length++;
88     }
89
90     // Get entry at index
91     const ZWeakRefData& at(size_t index) const {
92         assert(index < _length, "index out of bounds: %ld (length: %
93             ld)", index, _length);
94         return _data[index];
95     }
96
97     // Current length
98     size_t length() const {
99         return _length;
100     }
101
102     // Whether the array is empty
103     bool is_empty() const {
104         return _length == 0;
105     }
106
107     // Current capacity
108     size_t capacity() const {
109         return _capacity;
110     }
111
112     void clear_and_reserve(size_t new_capacity) {
113         if (new_capacity == 0) {

```

```

113 // Special case for clearing to empty - free memory and
    reset
114 if (_data != nullptr) {
115     FreeHeap(_data);
116     _data = nullptr;
117 }
118 _length = 0;
119 _capacity = 0;
120 return;
121 }
122 _length = 0;
123 if (_data != nullptr) {
124     FreeHeap(_data);
125 }
126 _capacity = MAX2((size_t) 8, next_power_of_2(new_capacity -
    1));
127 _data = (ZWeakRefData*)AllocateHeap(_capacity *
    _ZWeakRefData_size, mtGC);
128 }
129
130 // Clear without deallocating
131 void clear() {
132     _length = 0;
133 }
134
135 // Reserve capacity without clearing
136 void reserve(size_t new_capacity) {
137     if (new_capacity > _capacity) {
138         grow(new_capacity);
139     }
140 }
141 };
142
143 #endif // SHARE_GC_Z_ZWEAKREFARRAY_HPP

```

B.2 Reference Processor

B.2.1 Interface

File: zReferenceProcessor.hpp

```

1 /*
2  * Copyright (c) 2015, 2024, Oracle and/or its affiliates. All
    rights reserved.
3  * DO NOT ALTER OR REMOVE COPYRIGHT NOTICES OR THIS FILE HEADER.
4  *
5  * This code is free software; you can redistribute it and/or
    modify it

```

```

6  * under the terms of the GNU General Public License version 2
   * only, as
7  * published by the Free Software Foundation.
8  *
9  * This code is distributed in the hope that it will be useful,
   * but WITHOUT
10 * ANY WARRANTY; without even the implied warranty of
   * MERCHANTABILITY or
11 * FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public
   * License
12 * version 2 for more details (a copy is included in the
   * LICENSE file that
13 * accompanied this code).
14 *
15 * You should have received a copy of the GNU General Public
   * License version
16 * 2 along with this work; if not, write to the Free Software
   * Foundation,
17 * Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA.
18 *
19 * Please contact Oracle, 500 Oracle Parkway, Redwood Shores,
   * CA 94065 USA
20 * or visit www.oracle.com if you need additional information
   * or have any
21 * questions.
22 */
23
24 #ifndef SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP
25 #define SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP
26
27 #include "gc/shared/referenceDiscoverer.hpp"
28 #include "gc/z/zAddress.hpp"
29 #include "gc/z/zWeakRefArray.hpp"
30 #include "gc/z/zValue.hpp"
31
32 class ConcurrentGCTimer;
33 class ReferencePolicy;
34 class ZWorkers;
35
36 class ZReferenceProcessor : public ReferenceDiscoverer {
37     friend class ZReferenceProcessorTask;
38
39 private:
40     static const size_t ReferenceTypeCount = REF_PHANTOM + 1;
41     typedef size_t Counters[ReferenceTypeCount];
42
43     ZWorkers* const _workers;
44     ReferencePolicy* _soft_reference_policy;
45     bool _uses_clear_all_soft_reference_policy;
46     ZPerWorker<Counters> _encountered_count;

```

```

47 ZPerWorker<Counters> _discovered_count;
48 ZPerWorker<Counters> _enqueued_count;
49 ZPerWorker<size_t> _encountered_weak_refs_without_queue_count
   ;
50 ZPerWorker<size_t> _discovered_weak_refs_without_queue_count;
51 ZPerWorker<size_t> _cleared_weak_refs_without_queue_count;
52 ZPerWorker<zaddress> _discovered_list;
53 ZContended<zaddress> _pending_list;
54 zaddress _pending_list_tail;
55 ZPerWorker<ZWeakRefArray>
   _discovered_weak_refs_without_queue_arr;
56 ZPerWorker<bool> _array_empty;
57 OopHandle _null_queue_handle;
58
59 bool is_inactive(zaddress reference, oop referent,
   ReferenceType type) const;
60 bool is_strongly_live(oop referent) const;
61 bool is_softly_live(zaddress reference, ReferenceType type)
   const;
62
63 bool should_discover(zaddress reference, ReferenceType type,
   oop referent) const;
64 bool try_make_inactive(zaddress reference, ReferenceType type
   ) const;
65 bool try_make_inactive_fast(const ZWeakRefData& data);
66
67 void discover(zaddress reference, ReferenceType type,
   zaddress referent);
68
69 void verify_empty() const;
70
71 void process_worker_discovered_list(zaddress discovered_list);
72
73 void process_worker_discovered_weak_refs_without_queue(
   ZWeakRefArray& weak_refs_without_queue);
74 void work();
75 void collect_statistics();
76
77 zaddress swap_pending_list(zaddress pending_list);
78
79 inline bool has_reference_queue(zaddress reference);
80
81 void initialize_null_queue_handle();
82
83 public:
84 ZReferenceProcessor(ZWorkers* workers);
85
86 void set_soft_reference_policy(bool clear_all_soft_references
   );
87 bool uses_clear_all_soft_reference_policy() const;

```

```

88 void reset_statistics();
89
90 virtual bool discover_reference(oop reference, ReferenceType
    type);
91 void process_references();
92 void enqueue_references();
93
94 void verify_pending_references();
95
96 void initializeResources();
97 };
98
99 #endif // SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP

```

B.2.2 Implementation Excerpts

The following excerpts from `zReferenceProcessor.cpp` show the constructor and per-worker storage setup, the discovery path with queue-less weak-reference routing, the processing paths, and the queue sentinel initialisation.

Constructor and per-worker storage:

```

1  _soft_reference_policy(nullptr),
2  _uses_clear_all_soft_reference_policy(false),
3  _encountered_count(),
4  _discovered_count(),
5  _enqueued_count(),
6  _encountered_weak_refs_without_queue_count(),
7  _discovered_weak_refs_without_queue_count(),
8  _cleared_weak_refs_without_queue_count(),
9  _discovered_list(zaddress::null),
10 _pending_list(zaddress::null),
11 _pending_list_tail(zaddress::null),
12 _discovered_weak_refs_without_queue_arr(),
13 _array_empty(),
14 _null_queue_handle() {
15
16 _array_empty.set_all(true);
17 log_info(gc, ref) ("Reference Processor created, Config: All
    optimisations");
18 }
19
20 void ZReferenceProcessor::set_soft_reference_policy(bool
    clear_all_soft_references) {
21     static AlwaysClearPolicy always_clear_policy;
22     static LRUMaxHeapPolicy lru_max_heap_policy;
23
24     _uses_clear_all_soft_reference_policy =
        clear_all_soft_references;
25

```



```

26  if (clear_all_soft_references) {
27      _soft_reference_policy = &always_clear_policy;
28  } else {
29      _soft_reference_policy = &lru_max_heap_policy;
30  }
31
32  _soft_reference_policy->setup();
33  }
34
35  bool ZReferenceProcessor::uses_clear_all_soft_reference_policy
36      () const {
37      return _uses_clear_all_soft_reference_policy;
38  }
39
40  bool ZReferenceProcessor::is_inactive(zaddress reference, oop
41      referent, ReferenceType type) const {

```

Discovery path with queue-less routing:

```

1
2  assert (ZHeap::heap()->is_old(reference), "Must be old");
3  assert (is_null(reference_discovered(reference)), "Already
4      discovered");
5
6  // Update statistics
7  if (type == REF_WEAK && !has_reference_queue(reference)) {
8      _discovered_weak_refs_without_queue_count.get()++;
9  }
10 else {
11     _discovered_count.get()[type]++;
12 }
13
14 if (type == REF_WEAK && !has_reference_queue(reference)) {
15     zpointer* const referent_addr =
16         reference_referent_addr_non_vol(reference);
17     zaddress* const discovered_addr = reference_discovered_addr(
18         reference);
19     const zpointer referent_value = *referent_addr;
20
21     // WeakReference with null ReferenceQueue - remember for
22     // special processing
23     ZWeakRefArray& weak_refs_without_queue =
24         _discovered_weak_refs_without_queue_arr.get();
25     ZWeakRefData data = {referent_addr, discovered_addr,
26         referent, referent_value};
27     weak_refs_without_queue.append(data);
28     _array_empty.set(false);
29     reference_set_discovered(reference, reference); // mark as
30         discovered
31 } else {
32     if (type == REF_FINAL) {

```

```

27 // Mark referent (and its reachable subgraph) finalizable.
    This avoids
28 // the problem of later having to mark those objects if
    the referent is
29 // still final reachable during processing.
30 volatile zpointer* const referent_addr =
    reference_referent_addr(reference);
31 ZBarrier::mark_barrier_on_old_oop_field(referent_addr, true
    /* finalizable */);
32 }
33
34 // Add reference to discovered list
35 zaddress* const list = _discovered_list.addr();
36 reference_set_discovered(reference, *list);
37 *list = reference;
38 }
39 }
40
41 bool ZReferenceProcessor::discover_reference(oop reference_obj,
    ReferenceType type) {
42 if (!RegisterReferences) {
43 // Reference processing disabled
44 return false;
45 }
46
47 log_trace(gc, ref) ("Encountered Reference: " PTR_FORMAT " (%s
    )", p2i(reference_obj), reference_type_name(type));
48
49 const zaddress reference = to_zaddress(reference_obj);
50
51 // Update statistics
52 if (type == REF_WEAK && !has_reference_queue(reference)) {
53 _encountered_weak_refs_without_queue_count.get()++;
54 }
55 else {
56 _encountered_count.get()[type]++;
57 }
58
59 if (ZHeap::heap()->is_young(reference)) {
60 // Don't discover young references. Young gen reference
    processing is scary and therefore not supported.
61 return false;
62 }
63
64 volatile zpointer* const referent_addr =
    reference_referent_addr(reference);
65 const zaddress ref_raw_addr = ZBarrier::
    load_barrier_on_oop_field(referent_addr);
66 const oop referent = to_oop(ref_raw_addr);
67
68 if (!should_discover(reference, type, referent)) {

```

```

69     // Not discovered
70     return false;
71 }
72
73 discover(reference, type, ref_raw_addr);
74
75 // Discovered
76 return true;
77 }
78
79 void ZReferenceProcessor::process_worker_discovered_list(
    zaddress discovered_list) {
80     zaddress keep_head = zaddress::null;
81     zaddress keep_tail = zaddress::null;
82
83     // Iterate over the discovered list and unlink them as we go,
        potentially

```

Processing paths for separate discovered lists and dynamic arrays:

```

1     *((zpointer*)data.discovered_field_addr) = color_null(); //
        Mark as dropped
2     if (try_make_inactive_fast(data)) {
3         log_trace(gc, ref) ("\"Enqueued\" Weak Reference without
        Queue");
4         // Update statistics
5         _cleared_weak_refs_without_queue_count.get()++;
6     } else {
7         log_trace(gc, ref) ("Dropped Weak Reference without Queue");
8
9         dropped++;
10    }
11    SuspendibleThreadSet::yield();
12    weak_refs_without_queue.clear_and_reserve(dropped);
13 }
14
15 void ZReferenceProcessor::work() {
16     SuspendibleThreadSetJoiner sts_joiner;
17
18     ZPerWorkerIterator<zaddress> iter(&_discovered_list);
19     ZPerWorkerIterator<ZWeakRefArray> iter_weak_refs(&
        _discovered_weak_refs_without_queue_arr);
20     ZPerWorkerIterator<bool> iter_array_empty(&_array_empty);
21
22     zaddress* list_addr = nullptr;
23     ZWeakRefArray* array_addr = nullptr;
24     bool* array_empty = nullptr;
25
26     for (; iter.next(&list_addr) && iter_weak_refs.next(&
        array_addr) && iter_array_empty.next(&array_empty);) {
27

```

```

28     const zaddress discovered_list = AtomicAccess::xchg(
29         list_addr, zaddress::null);
30     const bool has_array = !AtomicAccess::xchg(array_empty, true
31         );
32     const bool has_discovered = discovered_list != zaddress::
33         null;
34
35     if (has_array) {
36         // Process discovered weak references without queue
37         process_worker_discovered_weak_refs_without_queue(*
38             array_addr);
39     }
40     if (has_discovered) {
41         // Process discovered references
42         process_worker_discovered_list(discovered_list);
43     }
44 }
45
46 void ZReferenceProcessor::verify_empty() const {
47     #ifndef ASSERT
48     ZPerWorkerConstIterator<zaddress> iter(&_discovered_list);
49     for (const zaddress* list; iter.next(&list);) {
50         assert(is_null(*list), "Discovered list not empty");
51     }
52
53     ZPerWorkerConstIterator<ZWeakRefArray> iter_weak_refs(&
54         _discovered_weak_refs_without_queue_arr);
55     for (const ZWeakRefArray* array; iter_weak_refs.next(&array)
56         ;) {
57         assert(array->is_empty(), "Discovered weak refs without
58             queue not empty");
59     }
60
61     assert(is_null(_pending_list.get()), "Pending list not empty"
62         );
63 #endif
64 }

```

Queue sentinel initialisation and queue checks:

```

1     return true;
2 }
3
4     const zaddress ref_queue_addr_value = ZBarrier::
5         load_barrier_on_oop_field(reference_queue_addr(reference));
6     const oop ref_queue = to_oop(ref_queue_addr_value);
7     bool result = ref_queue != _null_queue_handle.resolve();
8     return result;
9 }
10 void ZReferenceProcessor::initialize_null_queue_handle() {

```

```

11 EXCEPTION_MARK;
12 TempNewSymbol class_name = SymbolTable::new_symbol("java/lang
   /ref/ReferenceQueue");
13 Klass* k = SystemDictionary::resolve_or_fail(class_name, true,
   CHECK);
14 InstanceKlass* ik = InstanceKlass::cast(k);
15 ik->initialize(CHECK);
16 fieldDescriptor fd;
17 bool found = ik->find_local_field(SymbolTable::new_symbol("
   NULL_QUEUE"),
18                                     vmSymbols::referencequeue_signature
   (), &fd);
19 assert(found && fd.is_static(), "ReferenceQueue.NULL_QUEUE
   missing");
20 oop null_q = ik->java_mirror()->obj_field(fd.offset());
21 _null_queue_handle = OopHandle(Universe::vm_global(), null_q);
22 }
23
24 void ZReferenceProcessor::initializeResources() {
25   Thread* const current = Thread::current();
26   guarantee(current->is_Java_thread(), "Must be called by a
   JavaThread");
27   guarantee(!current->is_ConcurrentGC_thread(), "Must not be
   called by a GC thread");
28
29   initialize_null_queue_handle();
30 }

```

B.3 Weak Fields

B.3.1 Annotation

File: java/lang/ref/weak.java

```

1  /*
2   * Copyright (c) 2026, Oracle and/or its affiliates. All rights
   reserved.
3   * DO NOT ALTER OR REMOVE COPYRIGHT NOTICES OR THIS FILE HEADER.
4
5   *
6   * This code is free software; you can redistribute it and/or
   modify it
7   * under the terms of the GNU General Public License version 2
   only, as
8   * published by the Free Software Foundation. Oracle designates
   this
9   * particular file as subject to the "Classpath" exception as
   provided
   * by Oracle in the LICENSE file that accompanied this code.

```

```

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    but WITHOUT
12  * ANY WARRANTY; without even the implied warranty of
    MERCHANTABILITY or
13  * FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public
    License
14  * version 2 for more details (a copy is included in the
    LICENSE file that
15  * accompanied this code).
16  *
17  * You should have received a copy of the GNU General Public
    License version
18  * 2 along with this work; if not, write to the Free Software
    Foundation,
19  * Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA.
20  *
21  * Please contact Oracle, 500 Oracle Parkway, Redwood Shores,
    CA 94065 USA
22  * or visit www.oracle.com if you need additional information
    or have any
23  * questions.
24  */
25
26 package java.lang.ref;
27
28 import java.lang.annotation.*;
29
30 /**
31  * A field annotation for references that should be treated as
    weak by the VM.
32  */
33 @Retention(RetentionPolicy.RUNTIME)
34 @Target(ElementType.FIELD)
35 public @interface weak {}

```

B.3.2 Data Structure

File: zWeakFieldArray.hpp

```

1  /*
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    reserved.
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    modify it

```

```

6  * under the terms of the GNU General Public License version 2
   * only, as
7  * published by the Free Software Foundation.
8  *
9  * This code is distributed in the hope that it will be useful,
   * but WITHOUT
10 * ANY WARRANTY; without even the implied warranty of
   * MERCHANTABILITY or
11 * FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public
   * License
12 * version 2 for more details (a copy is included in the
   * LICENSE file that
13 * accompanied this code).
14 *
15 * You should have received a copy of the GNU General Public
   * License version
16 * 2 along with this work; if not, write to the Free Software
   * Foundation,
17 * Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA.
18 *
19 * Please contact Oracle, 500 Oracle Parkway, Redwood Shores,
   * CA 94065 USA
20 * or visit www.oracle.com if you need additional information
   * or have any
21 * questions.
22 */
23
24 #ifndef SHARE_GC_Z_ZWEAKFIELDARRAY_HPP
25 #define SHARE_GC_Z_ZWEAKFIELDARRAY_HPP
26
27 #include "gc/z/zAddress.hpp"
28 #include "memory/allocation.hpp"
29 #include "utilities/debug.hpp"
30 #include "utilities/powerOfTwo.hpp"
31 #include "utilities/globalDefinitions.hpp"
32 #include <string.h>
33
34 struct ZWeakFieldData {
35     volatile zpointer* field_addr;
36     zpointer field_value;
37 };
38
39 // High-performance growable array specifically for storing
   * discovered reference data.
40 // Uses array-of-structs (AoS) layout for better cache locality
   * when processing sequentially.
41 // The stored entry type is configured by the template parameter
   * .
42 //
43 // Performance optimizations:

```

```

44 // - Uses memcpy for bulk copying instead of element-by-element
    loops
45 // - Does not zero newly allocated memory (caller responsible
    for initialization)
46 // - Single clear_and_reserve operation to avoid separate clear
    /reserve calls
47 // - Array-of-structs layout improves cache efficiency for
    sequential processing
48 class ZWeakFieldArray : public AnyObj {
49 private:
50     ZWeakFieldData* _data;
51     const size_t _ZWeakFieldData_size = sizeof(ZWeakFieldData);
52     size_t _length;
53     size_t _capacity;
54
55     void expand_to(size_t new_capacity) {
56         assert(new_capacity > _capacity, "expected growth but %ld <=
            %ld", new_capacity, _capacity);
57
58         ZWeakFieldData* new_data = (ZWeakFieldData*)AllocateHeap(
            new_capacity * _ZWeakFieldData_size, mtGC);
59
60         if (_length > 0) {
61             // Use memcpy for trivially copyable types - much faster
            than element-by-element copy
62             memcpy(new_data, _data, _length * _ZWeakFieldData_size);
63         }
64
65         if (_data != nullptr) {
66             FreeHeap(_data);
67         }
68
69         _data = new_data;
70         _capacity = new_capacity;
71     }
72
73     void grow(size_t min_capacity) {
74         size_t new_capacity = MAX2((size_t) 8, next_power_of_2(
            min_capacity - 1));
75         expand_to(new_capacity);
76     }
77
78 public:
79     ZWeakFieldArray() : _data(nullptr), _length(0), _capacity(0)
        {}
80
81     ~ZWeakFieldArray() {
82         if (_data != nullptr) {
83             FreeHeap(_data);
84             _data = nullptr;
85         }

```



```

86     }
87
88     void append(const ZWeakFieldData& entry) {
89         if (_length >= _capacity) {
90             grow(_length + 1);
91         }
92         _data[_length] = entry;
93         _length++;
94     }
95
96     // Get entry at index
97     const ZWeakFieldData& at(size_t index) const {
98         assert(index < _length, "index out of bounds: %ld (length: %
99             ld)", index, _length);
100         return _data[index];
101     }
102
103     // Current length
104     size_t length() const {
105         return _length;
106     }
107
108     // Whether the array is empty
109     bool is_empty() const {
110         return _length == 0;
111     }
112
113     // Current capacity
114     size_t capacity() const {
115         return _capacity;
116     }
117
118     void clear_and_reserve(size_t new_capacity) {
119         if (new_capacity == 0) {
120             // Special case for clearing to empty - free memory and
121             // reset
122             if (_data != nullptr) {
123                 FreeHeap(_data);
124                 _data = nullptr;
125             }
126             _length = 0;
127             _capacity = 0;
128             return;
129         }
130         _length = 0;
131         if (_data != nullptr) {
132             FreeHeap(_data);
133         }
134         _capacity = MAX2((size_t) 8, next_power_of_2(new_capacity -
135             1));

```

```

133     _data = (ZWeakFieldData*)AllocateHeap(_capacity *
134         _ZWeakFieldData_size, mtGC);
135 }
136
137 // Clear without deallocating
138 void clear() {
139     _length = 0;
140 }
141
142 // Reserve capacity without clearing
143 void reserve(size_t new_capacity) {
144     if (new_capacity > _capacity) {
145         grow(new_capacity);
146     }
147 };
148
149 #endif // SHARE_GC_Z_ZWEAKFIELDARRAY_HPP

```

B.3.3 Reference Processor

File: zReferenceProcessor.hpp

```

1  /*
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3   * rights reserved.
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11  * published by the Free Software Foundation.
12  *
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16  * MERCHANTABILITY or
17  * FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public
18  * License
19  * version 2 for more details (a copy is included in the
20  * LICENSE file that
21  * accompanied this code).
22  *
23  * You should have received a copy of the GNU General Public
24  * License version
25  * 2 along with this work; if not, write to the Free Software
26  * Foundation,

```

```

17  * Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA.
18
19  * Please contact Oracle, 500 Oracle Parkway, Redwood Shores,
    CA 94065 USA
20  * or visit www.oracle.com if you need additional information
    or have any
21  * questions.
22  */
23
24  #ifndef SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP
25  #define SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP
26
27  #include "gc/shared/referenceDiscoverer.hpp"
28  #include "gc/z/zAddress.hpp"
29  #include "gc/z/zWeakFieldArray.hpp"
30  #include "gc/z/zValue.hpp"
31
32  class ConcurrentGCTimer;
33  class ReferencePolicy;
34  class ZWorkers;
35
36  class ZReferenceProcessor : public ReferenceDiscoverer {
37  friend class ZReferenceProcessorTask;
38
39  private:
40  static const size_t ReferenceTypeCount = REF_PHANTOM + 1;
41  typedef size_t Counters[ReferenceTypeCount];
42
43  ZWorkers* const _workers;
44  ReferencePolicy* _soft_reference_policy;
45  bool _uses_clear_all_soft_reference_policy;
46  ZPerWorker<Counters> _encountered_count;
47  ZPerWorker<Counters> _discovered_count;
48  ZPerWorker<Counters> _enqueued_count;
49  ZPerWorker<size_t> _encountered_weak_fields_count;
50  ZPerWorker<size_t> _discovered_weak_fields_count;
51  ZPerWorker<size_t> _cleared_weak_fields_count;
52  ZPerWorker<zaddress> _discovered_list;
53  ZContended<zaddress> _pending_list;
54  zaddress _pending_list_tail;
55  ZPerWorker<ZWeakFieldArray> _discovered_weak_fields;
56  ZPerWorker<bool> _discovered_weak_fields_empty;
57
58  bool is_inactive(zaddress reference, oop referent,
    ReferenceType type) const;
59  bool is_strongly_live(oop referent) const;
60  bool is_softly_live(zaddress reference, ReferenceType type)
    const;
61

```

```

62  bool should_discover(zaddress reference, ReferenceType type)
        const;
63  bool should_discover_weak_field(zpointer field_value) const;
64  bool try_make_inactive(zaddress reference, ReferenceType type
        ) const;
65
66  void discover(zaddress reference, ReferenceType type);
67
68  void verify_empty() const;
69
70  void process_worker_discovered_list(zaddress discovered_list);
71
72  void process_worker_discovered_weak_fields(ZWeakFieldArray&
        weak_fields);
72  void work();
73  void collect_statistics();
74
75  zaddress swap_pending_list(zaddress pending_list);
76
77 public:
78     ZReferenceProcessor(ZWorkers* workers);
79
80     bool discover_weak_field(volatile zpointer* field_addr);
81
82     void set_soft_reference_policy(bool clear_all_soft_references
        );
83     bool uses_clear_all_soft_reference_policy() const;
84
85     void reset_statistics();
86
87     virtual bool discover_reference(oop reference, ReferenceType
        type);
88     void process_references();
89     void enqueue_references();
90
91     void verify_pending_references();
92 };
93
94 #endif // SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP

```

The following excerpts from `zReferenceProcessor.cpp` show the discovery filtering, the discovery function, and the processing function.

Discovery filtering:

```

1  bool ZReferenceProcessor::should_discover_weak_field(zpointer
        field_value) const {
2      if (is_null_any(field_value)) {
3          // Field is already null, no need to discover
4          return false;
5      }
6

```

```

7  if (ZPointer::is_mark_good(field_value)) {
8      // Field points to a strongly live object, no need to
        discover
9      return false;
10 }
11
12 // Field points to an object that is not strongly live, we
        should discover
13 return true;
14 }

```

Weak-field discovery:

```

1  if (!should_discover_weak_field(field_value)) {
2      // Not discovered
3      return false;
4  }
5
6  // Discovered
7  log_trace(gc, ref) ("Discovered Weak Reference Field: "
        PTR_FORMAT, p2i(field_addr));
8
9  // Update statistics
10 _discovered_weak_fields_count.get()++;
11
12 ZWeakFieldArray& weak_fields = _discovered_weak_fields.get();
13 ZWeakFieldData data = {field_addr, field_value};
14 weak_fields.append(data);
15 _discovered_weak_fields_empty.set(false);
16
17 return true;
18 }
19
20 void ZReferenceProcessor::process_worker_discovered_list(
        zaddress discovered_list) {
21     zaddress keep_head = zaddress::null;
22     zaddress keep_tail = zaddress::null;
23
24     // Iterate over the discovered list and unlink them as we go,
        potentially
25     // appending them to the keep list
26     for (zaddress reference = discovered_list; !is_null(reference)
        ); ) {
27         assert (ZHeap::heap()->is_old(reference), "Must be old");
28
29         const ReferenceType type = reference_type(reference);
30         const zaddress next = reference_discovered(reference);
31         reference_set_discovered(reference, zaddress::null);
32
33         if (try_make_inactive(reference, type)) {
34             // Keep reference

```

```

35     log_trace(gc, ref)("Enqueued Reference: " PTR_FORMAT " (%s
        ), untype(reference), reference_type_name(type));

```

Weak-field processing:

```

1  void ZReferenceProcessor::process_worker_discovered_weak_fields(
    ZWeakFieldArray& weak_fields) {
2      size_t dropped = 0;
3      for (size_t i = 0; i < weak_fields.length(); i++) {
4          const ZWeakFieldData& data = weak_fields.at(i);
5          if (ZBarrier::clean_barrier_on_weak_oop_field_preloaded(data.
                field_addr, data.field_value)) {
6              log_trace(gc, ref)("\nCleared" Weak Reference Field: "
                    PTR_FORMAT, p2i(data.field_addr));
7              // Update statistics
8              _cleared_weak_fields_count.get()++;
9          } else {
10             log_trace(gc, ref)("Dropped Weak Reference Field: "
                PTR_FORMAT, p2i(data.field_addr));
11             dropped++;
12         }
13         SuspendibleThreadSet::yield();
14     }
15     weak_fields.clear_and_reserve(dropped);
16 }
17
18 void ZReferenceProcessor::work() {
19     SuspendibleThreadSetJoiner sts_joiner;
20
21     ZPerWorkerIterator<ZWeakFieldArray> iter_weak_fields(&
        _discovered_weak_fields);
22     ZPerWorkerIterator<bool> iter_weak_fields_empty(&
        _discovered_weak_fields_empty);
23
24     ZWeakFieldArray* weak_fields_addr = nullptr;
25     bool* weak_fields_empty = nullptr;
26
27     for (; iter_weak_fields.next(&weak_fields_addr) &&
        iter_weak_fields_empty.next(&weak_fields_empty);) {
28         const bool has_weak_fields = !AtomicAccess::xchg(
            weak_fields_empty, true);
29         if (has_weak_fields) {
30             process_worker_discovered_weak_fields(*weak_fields_addr);
31         }
32     }
33
34     ZPerWorkerIterator<zaddress> iter(&_discovered_list);
35     for (zaddress* start; iter.next(&start);) {
36         const zaddress discovered_list = AtomicAccess::xchg(start,
            zaddress::null);
37
38         if (discovered_list != zaddress::null) {

```

```

39     // Process discovered references
40     process_worker_discovered_list(discovered_list);
41 }

```

B.3.4 Marking Closure

The following excerpt from `zMark.cpp` shows how weak-annotated fields are detected and diverted to the weak-field discovery path during object marking.

Field annotation check and discovery diversion:

```

1  bool is_weak_annotated_field(oop* p) const {
2      if (_current_instance_klass == nullptr) {
3          return false;
4      }
5
6      const uintptr_t addr = (uintptr_t)p;
7      if (addr < _current_obj_start || addr >= _current_obj_end) {
8          return false;
9      }
10
11     const int offset = (int)(addr - _current_obj_start);
12     fieldDescriptor fd;
13     if (!_current_instance_klass->find_field_from_offset(offset,
14         false /* is_static */, &fd)) {
15         return false;
16     }
17     return fd.is_weak();
18 }
19
20 const bool _visit_metadata;
21
22 public:
23     ZMarkBarrierFollowOopClosure()
24         : OopIterateClosure(discoverer()),
25           _current_instance_klass(nullptr),
26           _current_obj_start(0),
27           _current_obj_end(0),
28           _visit_metadata(visit_metadata()) {}
29
30     void set_iteration_object(oop obj) {
31         if (obj->klass()->is_instance_klass()) {
32             _current_instance_klass = InstanceKlass::cast(obj->klass())
33                 );
34             _current_obj_start = cast_from_oop<uintptr_t>(obj);
35             _current_obj_end = _current_obj_start + (obj->size() *
36                 HeapWordSize);
37         } else {
38             _current_instance_klass = nullptr;
39         }
40     }

```

```

37     _current_obj_start = 0;
38     _current_obj_end = 0;
39 }
40 }
41
42 void clear_iteration_object() {
43     _current_instance_klass = nullptr;
44     _current_obj_start = 0;
45     _current_obj_end = 0;
46 }
47
48 virtual void do_oop(oop* p) {
49     if (!finalizable && is_weak_annotated_field(p)) {
50         if (ZGeneration::old()->discover_weak_field((volatile
51             zpointer*)p)) {
52             return;
53         }
54     }
55
56     switch (generation) {
57     case ZGenerationIdOptional::young:
58         ZBarrier::mark_barrier_on_young_oop_field((volatile
59             zpointer*)p);
60         break;
61     case ZGenerationIdOptional::old:
62         ZBarrier::mark_barrier_on_old_oop_field((volatile zpointer
63             *)p, finalizable);
64         break;
65     case ZGenerationIdOptional::none:
66         ZBarrier::mark_barrier_on_oop_field((volatile zpointer*)p,
67             finalizable);
68         break;
69     }
70 }

```

B.3.5 Compiler and Interpreter Support

The following excerpts show how the @weak decorator is applied in each execution tier. In all three tiers, the pattern is the same: if the field carries the weak flag and ZGC is active, ON_WEAK_OOP_REF is added to the decorator set before the access is performed.

C2 (opto/parse3.cpp) – field load and store:

```

1 void Parse::do_get_xxx(Node* obj, ciField* field, bool is_field)
2 {
3     BasicType bt = field->layout_type();
4     // Does this field have a constant value? If so, just push the
5     // value.

```



```

5  if (field->is_constant() &&
6      // Keep consistent with types found by ciTypeFlow: for an
7      // unloaded field type, ciTypeFlow::StateVector::
8      // do_getstatic()
9      // speculates the field is null. The code in the rest of
10     this
11     // method does the same. We must not bypass it and use a
12     non
13     // null constant here.
14     (bt != T_OBJECT || field->type()->is_loaded())) {
15     // final or stable field
16     Node* con = make_constant_from_field(field, obj);
17     if (con != nullptr) {
18         push_node(field->layout_type(), con);
19         return;
20     }
21 }

22 ciType* field_klass = field->type();
23 bool is_vol = field->is_volatile();
24
25 // Compute address and memory type.
26 int offset = field->offset_in_bytes();
27 const TypePtr* adr_type = C->alias_type(field)->adr_type();
28 Node *adr = basic_plus_adr(obj, obj, offset);
29 assert(C->get_alias_index(adr_type) == C->get_alias_index(
30     _gvn.type(adr)->isa_ptr()),
31     "slice of address and input slice don't match");
32
33 // Build the resultant type of the load
34 const Type *type;
35
36 bool must_assert_null = false;
37
38 DecoratorSet decorators = IN_HEAP;
39 decorators |= is_vol ? MO_SEQ_CST : MO_UNORDERED;
40
41 bool is_obj = is_reference_type(bt);
42
43 if (is_obj) {
44     if (UseZGC && field->is_weak()) {
45         decorators |= ON_WEAK_OOP_REF;
46     }
47     if (!field->type()->is_loaded()) {
48         type = TypeInstPtr::BOTTOM;
49         must_assert_null = true;
50     } else if (field->is_static_constant()) {
51         // This can happen if the constant oop is non-perm.
52         ciObject* con = field->constant_value().as_object();
53         // Do not "join" in the previous type; it doesn't add
54         value,

```

```

51 // and may yield a vacuous result if the field is of
    interface type.
52 if (con->is_null_object()) {
53     type = TypePtr::NULL_PTR;
54 } else {
55     type = TypeOopPtr::make_from_constant(con->isa_oopptr());
56 }
57 assert(type != nullptr, "field singleton type must be
    consistent");
58 } else {
59     type = TypeOopPtr::make_from_class(field_class->as_klass());
60 }
61 } else {
62     type = Type::get_const_basic_type(bt);
63 }
64
65 Node* ld = access_load_at(obj, adr, adr_type, type, bt,
    decorators);
66 if (is_obj && UseZGC && field->is_weak()) {
67     insert_mem_bar(Op_MemBarCPUOrder);
68 }
69
70 // Adjust Java stack
71 if (type2size[bt] == 1)
72     push(ld);
73 else
74     push_pair(ld);
75
76 if (must_assert_null) {
77     // Do not take a trap here. It's possible that the program
78     // will never load the field's class, and will happily see
79     // null values in this field forever. Don't stumble into a
80     // trap for such a program, or we might get a long series
81     // of useless recompilations. (Or, we might load a class
82     // which should not be loaded.) If we ever see a non-null
83     // value, we will then trap and recompile. (The trap will
84     // not need to mention the class index, since the class will
85     // already have been loaded if we ever see a non-null value
86     .)
87     // uncommon_trap(iter().get_field_signature_index());
88     if (PrintOpto && (Verbose || WizardMode)) {
89         method()->print_name(); tty->print_cr(" asserting nullness
            of field at bci: %d", bci());
90     }
91     if (C->log() != nullptr) {
92         C->log()->elem("assert_null reason='field' klass='%d'",
            C->log()->identify(field->type()));
93     }
94     // If there is going to be a trap, put it at the next
        bytecode:

```

```

95     set_bci(iter().next_bci());
96     null_assert(peek());
97     set_bci(iter().cur_bci()); // put it back
98 }
99 }
100
101 void Parse::do_put_xxx(Node* obj, ciField* field, bool is_field)
102 {
103     bool is_vol = field->is_volatile();
104     // Compute address and memory type.
105     int offset = field->offset_in_bytes();
106     const TypePtr* adr_type = C->alias_type(field)->adr_type();
107     Node* adr = basic_plus_adr(obj, obj, offset);
108     assert(C->get_alias_index(adr_type) == C->get_alias_index(
109         _gvn.type(adr)->isa_ptr()),
110         "slice of address and input slice don't match");
111     BasicType bt = field->layout_type();
112     // Value to be stored
113     Node* val = type2size[bt] == 1 ? pop() : pop_pair();
114     DecoratorSet decorators = IN_HEAP;
115     decorators |= is_vol ? MO_SEQ_CST : MO_UNORDERED;
116
117     bool is_obj = is_reference_type(bt);
118
119     if (is_obj) {
120         if (UseZGC && field->is_weak()) {
121             decorators |= ON_WEAK_OOP_REF;
122         }
123     }
124
125     // Store the value.
126     const Type* field_type;
127     if (!field->type()->is_loaded()) {
128         field_type = TypeInstPtr::BOTTOM;
129     } else {
130         if (is_obj) {
131             field_type = TypeOopPtr::make_from_klass(field->type()->
132                 as_klass());
133         } else {
134             field_type = Type::BOTTOM;
135         }
136     }
137     access_store_at(obj, adr, adr_type, val, field_type, bt,
138         decorators);
139
140     if (is_field) {
141         // Remember we wrote a volatile field.
142         // For not multiple copy atomic cpu (ppc64) a barrier should
143         be issued

```

```

141 // in constructors which have such stores. See do_exits() in
    parse1.cpp.
142 if (is_vol) {
143     set_wrote_volatile(true);
144 }
145 set_wrote_fields(true);
146
147 // If the field is final, the rules of Java say we are in <
    init> or <clinit>.
148 // If the field is @Stable, we can be in any method, but we
    only care about
149 // constructors at this point.
150 //
151 // Note the presence of writes to final/@Stable non-static
    fields, so that we
152 // can insert a memory barrier later on to keep the writes
    from floating
153 // out of the constructor.
154 if (field->is_final() || field->is_stable()) {

```

C1 (c1/c1_LIRGenerator.cpp) – field load decorator:

```

1 }
2
3 DecoratorSet decorators = IN_HEAP;
4 if (is_volatile) {
5     decorators |= MO_SEQ_CST;
6 }
7 if (needs_patching) {
8     decorators |= C1_NEEDS_PATCHING;
9 } else if (UseZGC && field_type == T_OBJECT && x->field()->
    is_weak()) {
10     decorators |= ON_WEAK_OOP_REF;
11 }
12
13 LIR_Opr result = rlock_result(x, field_type);
14 access_load_at(decorators, field_type,
15                 object, LIR_OprFact::intConst(x->offset()), result,
16                 info ? new CodeEmitInfo(info) : nullptr, info);
17 }

```

C1 (c1/c1_LIRGenerator.cpp) – field store decorator:

```

1 DecoratorSet decorators = IN_HEAP;
2 if (is_volatile) {
3     decorators |= MO_SEQ_CST;
4 }
5
6 if (needs_patching) {
7     decorators |= C1_NEEDS_PATCHING;
8 } else if (UseZGC && field_type == T_OBJECT && x->field()->
    is_weak()) {

```

```

9     decorators |= ON_WEAK_OOP_REF;
10 }
11
12 access_store_at(decorators, field_type, object, LIR_OprFact::
13     intConst(x->offset()),
14     value.result(), info != nullptr ? new
15         CodeEmitInfo(info) : nullptr, info);
16 }
17
18 void LIRGenerator::do_StoreIndexed(StoreIndexed* x) {
19     assert(x->is_pinned(), "");
20     bool needs_range_check = x->compute_needs_range_check();

```

Template Interpreter (cpu/x86/templateTable_x86.cpp) – getfield weak path:

```

1  __ bind(notBool);
2  __ cmpl(tos_state, atos);
3  __ jcc(Assembler::notEqual, notObj);
4  // atos
5  if (UseZGC) {
6  __ testl(flags, (1 << ResolvedFieldEntry::is_weak_shift));
7  __ jcc(Assembler::zero, strongOopLoad);
8  // Weak load path
9  do_oop_load(_masm, field, rax, ON_WEAK_OOP_REF);
10 __ push(atos);
11 // Don't rewrite bytecode for weak references, the fast path
12 // is only for strong loads
13 __ jmp(Done);
14 __ bind(strongOopLoad);
15 }
16 do_oop_load(_masm, field, rax);
17 __ push(atos);
18 if (!is_static && rc == may_rewrite) {
19     patch_bytecode(Bytecodes::_fast_agetfield, bc, rbx);
20 }
21 __ jmp(Done);

```

Template Interpreter (cpu/x86/templateTable_x86.cpp) – putfield weak path:

```

1  // atos
2  {
3  if (UseZGC) {
4  // Test weak flag before pop(atos) overwrites flags (rax)
5  __ testl(flags, (1 << ResolvedFieldEntry::is_weak_shift));
6  __ jcc(Assembler::notZero, strongOopStore);
7  // Weak store path
8  __ pop(atos);
9  if (!is_static) pop_and_check_object(obj);
10 do_oop_store(_masm, field, rax, ON_WEAK_OOP_REF);

```

```

11      // Don't rewrite bytecode for weak references, the fast
12      // path is only for strong stores
13      __ jmp(Done);
14      // Strong store path
15      __ bind(strongOopStore);
16  }
17
18  __ pop(atos);
19  if (!is_static) pop_and_check_object(obj);
20  do_oop_store(_masm, field, rax);
21  if (!is_static && rc == may_rewrite) {
22      patch_bytecode(Bytecodes::_fast_aputfield, bc, rbx, true,
23                      byte_no);
24  }
25  __ jmp(Done);
26  }

```

Appendix C.

Per-Variant Patch Details

The following tables list every modified or added file in each patch variant, together with its insertion and deletion counts relative to the upstream OpenJDK master branch. New files (not present in the upstream source) are listed with their total line count and no deletion count. For variants that use the shared base patch, its files are listed first under a “Base patch” heading within each table. The `dyn_only` and `weak_fields` variants do not include the base patch.

C.1 all

File	+	–
<i>Base patch</i>		
gc/shared/collectedHeap.hpp	+9	–3
gc/z/zCollectedHeap.cpp	+7	–0
gc/z/zCollectedHeap.hpp	+1	–0
gc/z/zGeneration.hpp	+6	–4
gc/z/zGeneration.inline.hpp	+3	–0
gc/z/zStat.cpp	+13	–8
gc/z/zStat.hpp	+2	–1
runtime/threads.cpp	+2	–0
<i>Variant-specific patch</i>		
gc/z/zReferenceProcessor.cpp	+184	–40
gc/z/zReferenceProcessor.hpp	+27	–12
gc/z/zWeakRefArray.hpp (new)	+143	—
Total	+397	–68

C.2 clear_path_dyn

File	+	–
<i>Base patch</i>		

File	+	−
gc/shared/collectedHeap.hpp	+9	−3
gc/z/zCollectedHeap.cpp	+7	−0
gc/z/zCollectedHeap.hpp	+1	−0
gc/z/zGeneration.hpp	+6	−4
gc/z/zGeneration.inline.hpp	+3	−0
gc/z/zStat.cpp	+13	−8
gc/z/zStat.hpp	+2	−1
runtime/threads.cpp	+2	−0
<i>Variant-specific patch</i>		
gc/z/zReferenceProcessor.cpp	+175	−55
gc/z/zReferenceProcessor.hpp	+26	−13
gc/z/zWeakRefArray.hpp (new)	+155	—
Total	+399	−84

C.3 clear_path_only

File	+	−
<i>Base patch</i>		
gc/shared/collectedHeap.hpp	+9	−3
gc/z/zCollectedHeap.cpp	+7	−0
gc/z/zCollectedHeap.hpp	+1	−0
gc/z/zGeneration.hpp	+6	−4
gc/z/zGeneration.inline.hpp	+3	−0
gc/z/zStat.cpp	+13	−8
gc/z/zStat.hpp	+2	−1
runtime/threads.cpp	+2	−0
<i>Variant-specific patch</i>		
gc/z/zReferenceProcessor.cpp	+129	−25
gc/z/zReferenceProcessor.hpp	+16	−0
Total	+188	−41

C.4 clear_path_sep

File	+	−
<i>Base patch</i>		
gc/shared/collectedHeap.hpp	+9	−3
gc/z/zCollectedHeap.cpp	+7	−0

File	+	−
gc/z/zCollectedHeap.hpp	+1	−0
gc/z/zGeneration.hpp	+6	−4
gc/z/zGeneration.inline.hpp	+3	−0
gc/z/zStat.cpp	+13	−8
gc/z/zStat.hpp	+2	−1
runtime/threads.cpp	+2	−0
<i>Variant-specific patch</i>		
gc/z/zReferenceProcessor.cpp	+153	−30
gc/z/zReferenceProcessor.hpp	+18	−0
Total	+214	−46

C.5 dyn_only

File	+	−
gc/z/zReferenceProcessor.cpp	+39	−56
gc/z/zReferenceProcessor.hpp	+17	−13
gc/z/zWeakRefArray.hpp (new)	+149	—
Total	+205	−69

C.6 sep_dyn

File	+	−
<i>Base patch</i>		
gc/shared/collectedHeap.hpp	+9	−3
gc/z/zCollectedHeap.cpp	+7	−0
gc/z/zCollectedHeap.hpp	+1	−0
gc/z/zGeneration.hpp	+6	−4
gc/z/zGeneration.inline.hpp	+3	−0
gc/z/zStat.cpp	+13	−8
gc/z/zStat.hpp	+2	−1
runtime/threads.cpp	+2	−0
<i>Variant-specific patch</i>		
gc/z/zReferenceProcessor.cpp	+144	−35
gc/z/zReferenceProcessor.hpp	+24	−10
gc/z/zWeakRefArray.hpp (new)	+149	—
Total	+360	−61

C.7 sep_only

File	+	−
<i>Base patch</i>		
gc/shared/collectedHeap.hpp	+9	−3
gc/z/zCollectedHeap.cpp	+7	−0
gc/z/zCollectedHeap.hpp	+1	−0
gc/z/zGeneration.hpp	+6	−4
gc/z/zGeneration.inline.hpp	+3	−0
gc/z/zStat.cpp	+13	−8
gc/z/zStat.hpp	+2	−1
runtime/threads.cpp	+2	−0
<i>Variant-specific patch</i>		
gc/z/zReferenceProcessor.cpp	+118	−27
gc/z/zReferenceProcessor.hpp	+12	−0
Total	+173	−43

C.8 weak_fields

File	+	−
cpu/x86/templateTable_x86.cpp	+34	−13
cpu/x86/templateTable_x86.hpp	+1	−1
cl/cl_LIRGenerator.cpp	+5	−0
cl/cl_Runtime1.cpp	+12	−3
ci/ciField.cpp	+7	−11
ci/ciField.hpp	+1	−0
ci/ciFlags.hpp	+5	−3
classfile/classFileParser.cpp	+42	−7
classfile/vmSymbols.hpp	+4	−7
gc/shared/accessBarrierSupport.cpp	+25	−6
gc/z/zBarrier.cpp	+39	−6
gc/z/zBarrier.hpp	+1	−0
gc/z/zBarrier.inline.hpp	+4	−0
gc/z/zGeneration.hpp	+6	−4
gc/z/zGeneration.inline.hpp	+3	−0
gc/z/zMark.cpp	+57	−8
gc/z/zReferenceProcessor.cpp	+110	−34
gc/z/zReferenceProcessor.hpp	+18	−12
gc/z/zStat.cpp	+12	−7
gc/z/zStat.hpp	+2	−1
gc/z/zWeakFieldArray.hpp (new)	+149	—

File	+	−
oops/fieldInfo.hpp	+5	−2
oops/resolvedFieldEntry.cpp	+32	−1
oops/resolvedFieldEntry.hpp	+17	−16
opto/library_call.cpp	+82	−77
opto/parse3.cpp	+11	−0
prims/jvmtiTagMap.cpp	+15	−8
runtime/fieldDescriptor.hpp	+1	−0
services/heapDumper.cpp	+13	−5
java.base/.../ref/weak.java (new)	+35	—
Total	+748	−232